Co-Designing Mechanisms and Motion: Towards Safe and Deterministic Human–Robot Interaction

Durgesh Haribhau Salunkhe

Requested host institutions:

1. ICube, Strasbourg: Team RDH
2. LAAS, Toulouse: Team Gepetto

Concours 03/02

An urgent need!

Blocs opératoires : la délégation d'actes infirmiers jusque-là réservés aux Ibode jette le trouble

PAR ARNAUD JANIN - PUBLIÉ LE 24/01/2025

0 RÉACTIONS COMMENTER

Le Quotidien du Médecin, 24 Jan 2025

Parution du décret permettant aux IDE, à titre transitoire et sur autorisation, la pratique des actes exclusifs IBODE

FHF FÉDÉRATION HOSPITALIÈRE DE FRANCE 03 Feb, 2025

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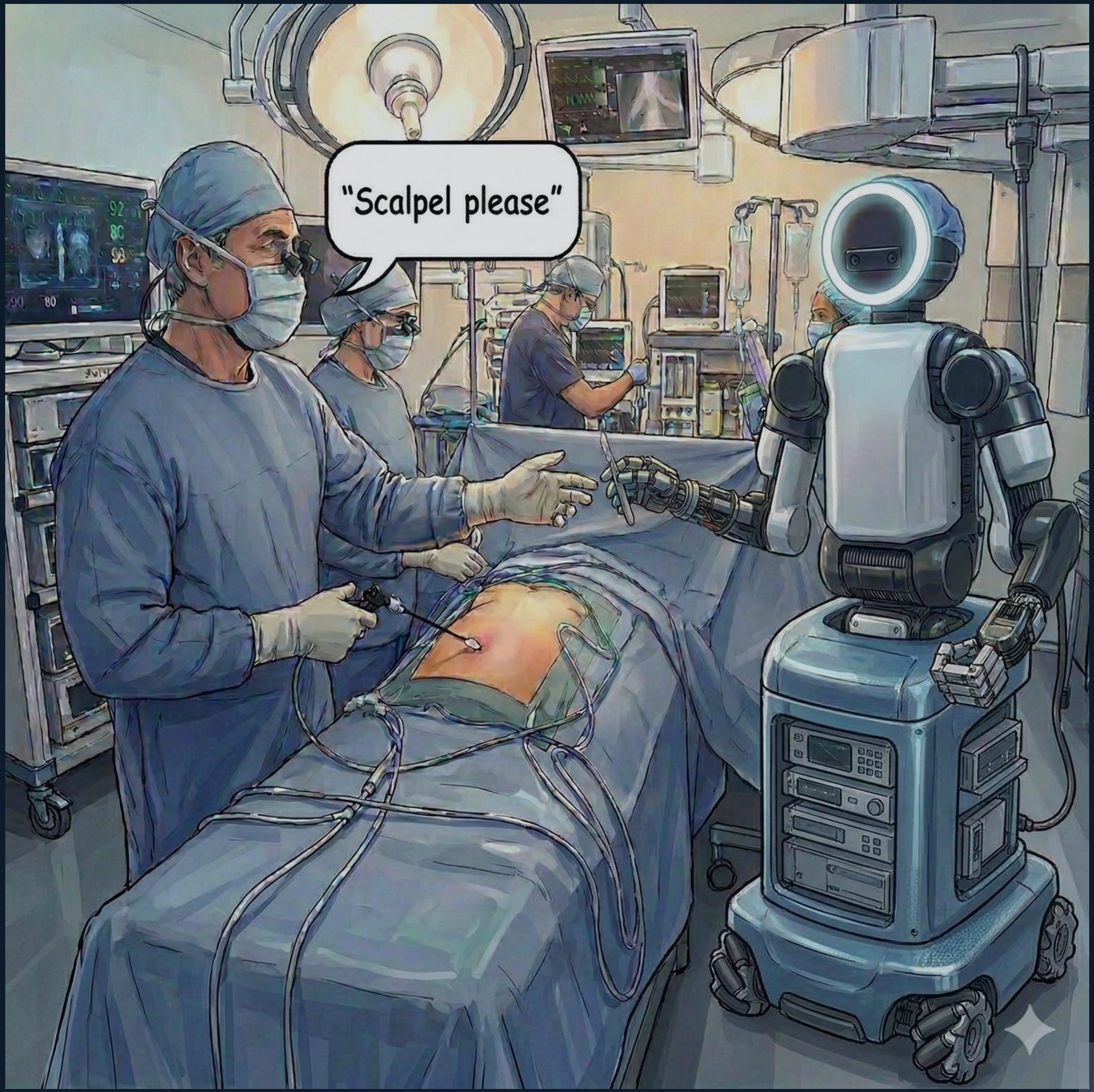
FHF FÉDÉRATION HOSPITALIÈRE DE FRANCE

03 Feb, 2025

Versius system (UK) used in Chemnitz Hospital, Germany

Please pass me the scalpel

An example scenario



AI-generated image

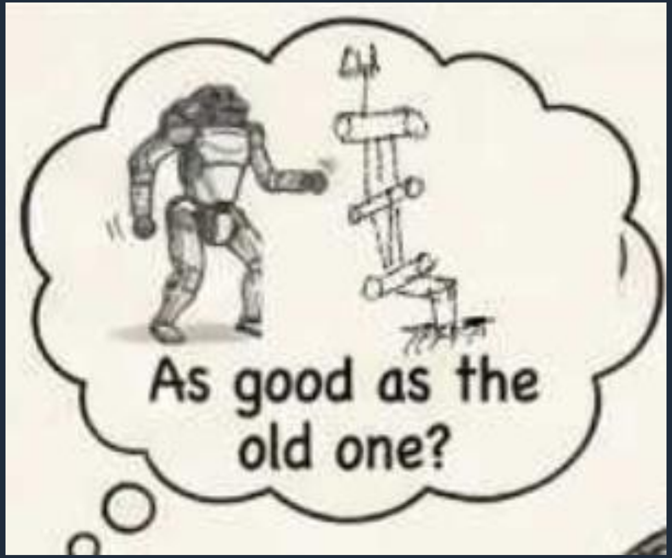
The three barriers to ubiquitous robots

Why a surgeon may hesitate to trust a robot

Reliable Execution

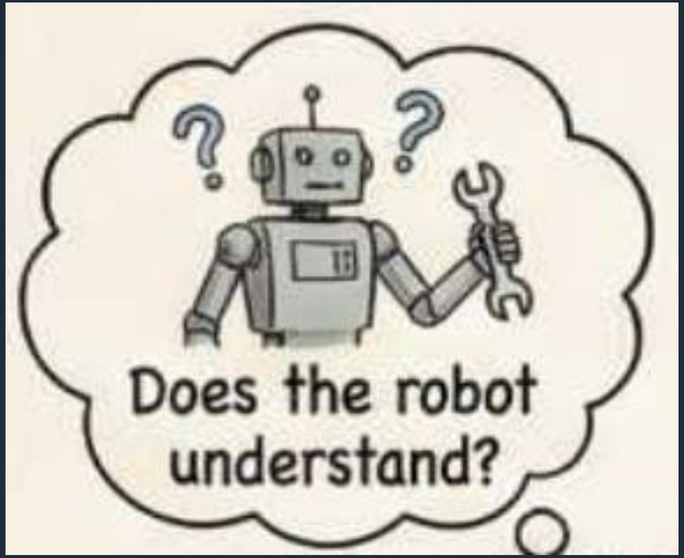
The real world is unstructured: robot execution must be safe and explainable.

Skill Scalability



As new robots are developed, skills need to safely transferred across all generations of robots.

Embodied alignment

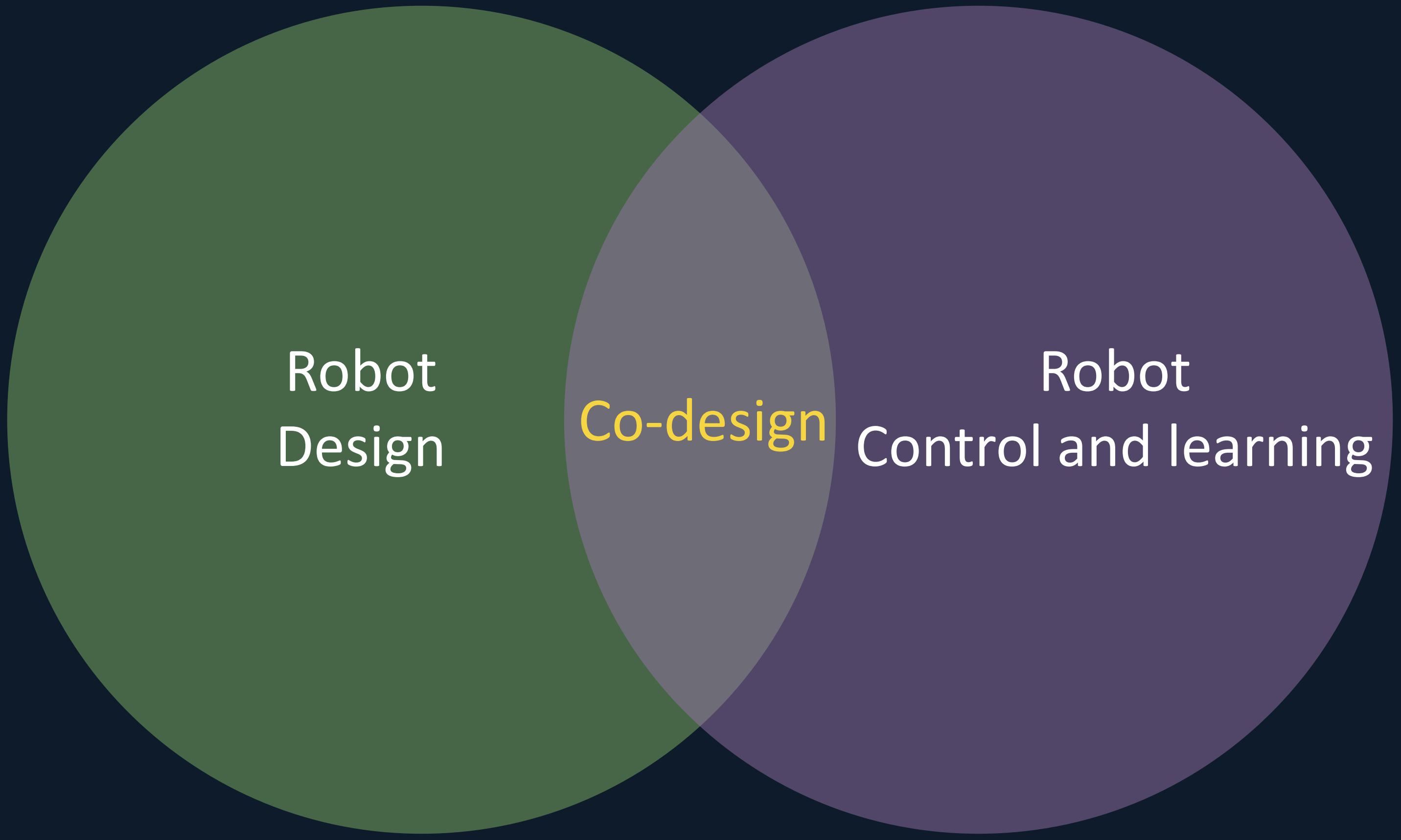


Robot design and actions should be kinematically-grounded while working with LLM and GenAI

Research Approach

Co-design approach: to develop robot designs and learning paradigms together, from the ground up, for human-centric tasks.

(Cheney et al., ’18)



Research timeline

Research Theme 1: **Task-Oriented Robot Design**

1. Design optimization of robots for otological surgery

Journals: MMT'20

Conferences: ICRA'22

Software: ParaOpt (used by LS2N, DFKI)

Research Theme 2: **Safe and Reliable Robot Behavior**

1. Active force compliance in 6-RSS parallel platform

Journals: RCIM'19

Conferences: AIR'20

Software:

2018-20

Università di Genova

CENTRALE NANTES

Director: Damien Chablat

Collaborations:
CHU Nantes, DFKI Germany

Research timeline

Research Theme 1: **Task-Oriented Robot Design**

1. Design optimization of robots for otological surgery
2. Kinematic analysis of generic 3R and 6R robots

Journals: MMT’20, MMT’22

Conferences: ICRA’22, ARK’22, ISSAC’22

Software: ParaOpt (used by LS2N, DFKI)

Research Theme 2: **Safe and Reliable Robot Behavior**

1. Active force compliance in 6-RSS parallel platform
2. Path planning framework for 6R robots

Journals: RCIM’19, IJRR’24

Conferences: AIR’20, ICRA’23, IROS’23

Software:

2018-20

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2020-23

Director: Philippe Wenger, Damien Chablat

Collaborations with:
Sorbonne, Linz, Innsbruck

Research timeline

Research Theme 1: **Task-Oriented Robot Design**

1. Design optimization of robots for otological surgery
2. Kinematic analysis of generic 3R and 6R robots
3. Novel designs for Human-Robot Interaction

Journals: MMT'20, MMT'22, RA-L'25

Conferences: ICRA'22, ARK'22, ISSAC'22, CK'25, IMA'25

Software: ParaOpt (used by LS2N, DFKI)

Research Theme 2: **Safe and Reliable Robot Behavior**

1. Active force compliance in 6-RSS parallel platform
2. Path planning framework for 6R robots
3. Kinematic Intelligence for cross-robot skill transfer

Journals: RCIM'19, IJRR'24, Science Robotics '26

Conferences: AIR'20, ICRA'23, IROS'23

Software: Skill Transfer (April '26)

2018-20

Università di Genova

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2020-23

LS2N

cnrs

Director: Philippe Wenger, Damien Chablat

Collaborations with:
Sorbonne, Linz, Innsbruck

2024-Current

EPFL

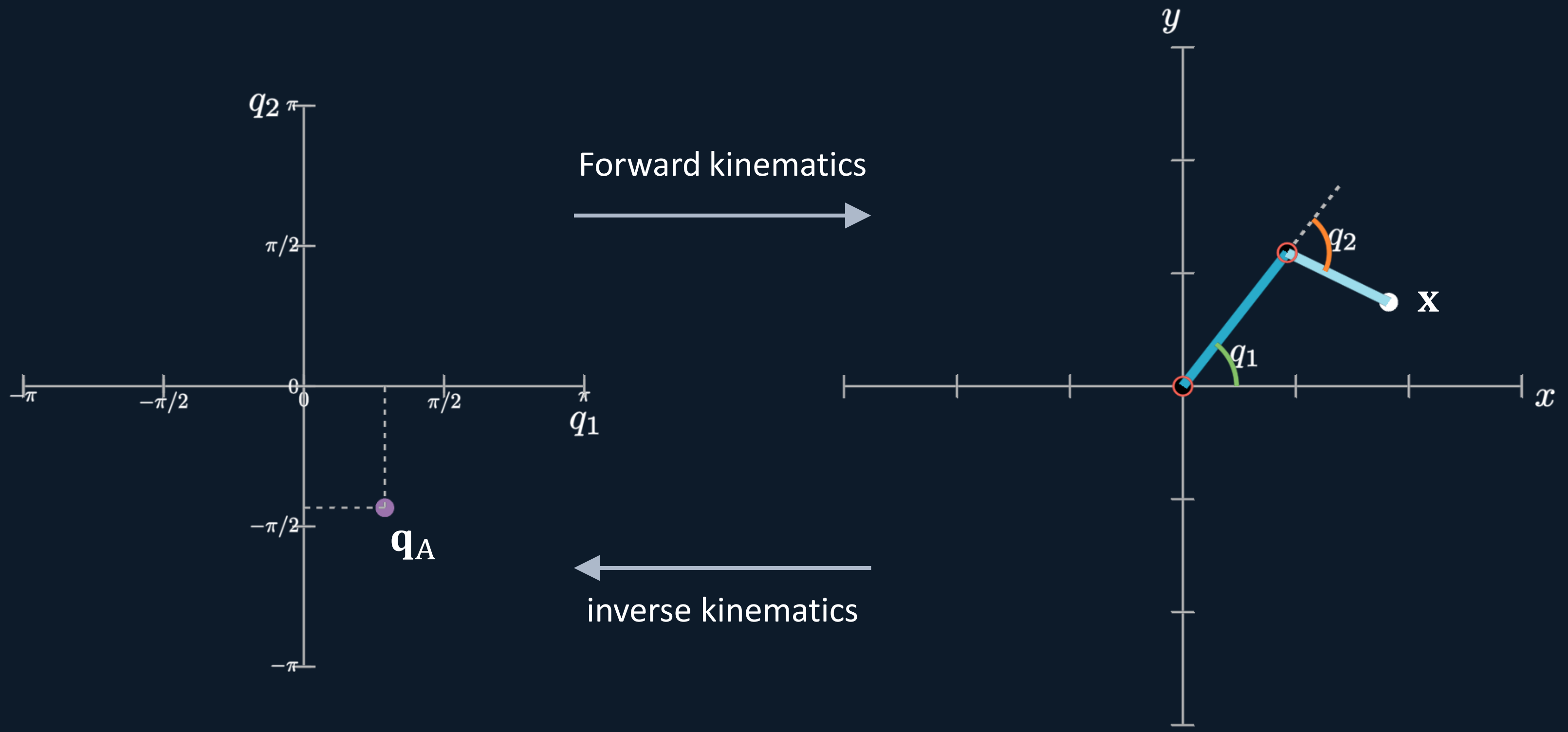
Supervisor: Aude Billard

Collaborations with:
IIT, India, IRI, Spain

Research activity

Doctoral work

nR robot: A robot with ‘n’ revolute joints. (I will use 2R, 3R, 6R and 7R)



Joint space

Workspace

Research activity

Doctoral work

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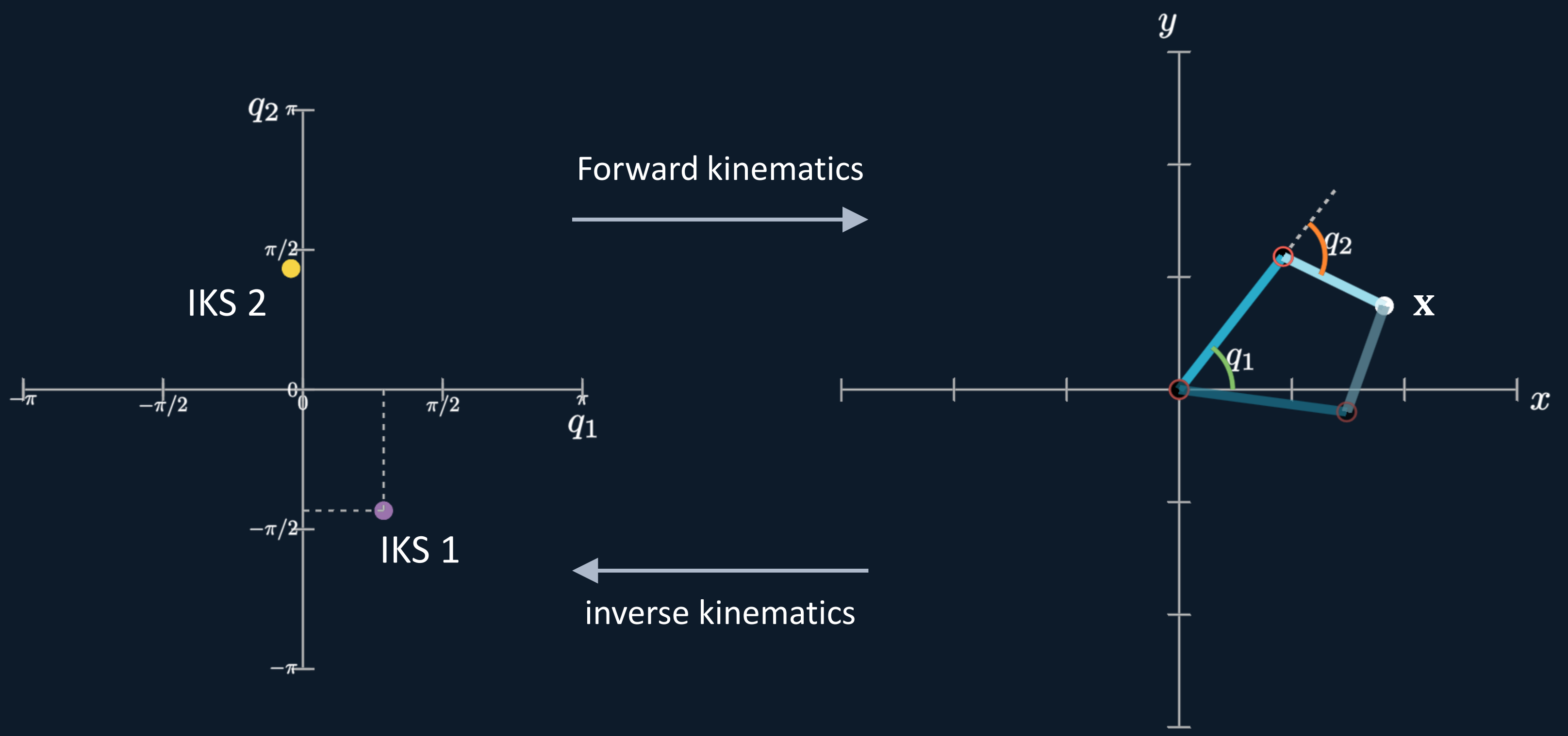
Inverse kinematic solutions(IKS): different numbers of ways a robot can reach the same target



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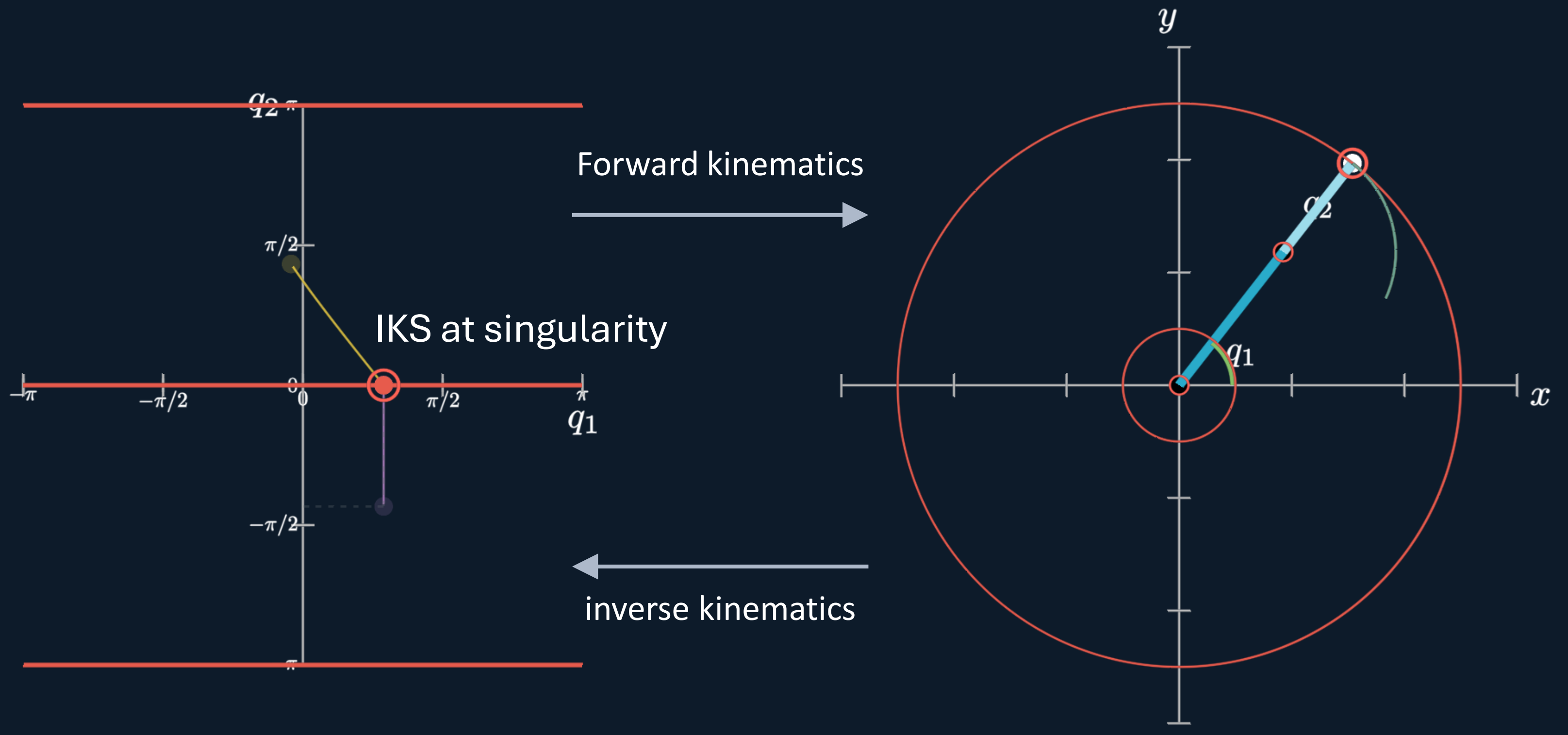
Singularity: Configurations where robot loses motion capabilities, control becomes difficult or unreliable



Joint space singularities

Workspace singularities

Research activity

Doctoral work

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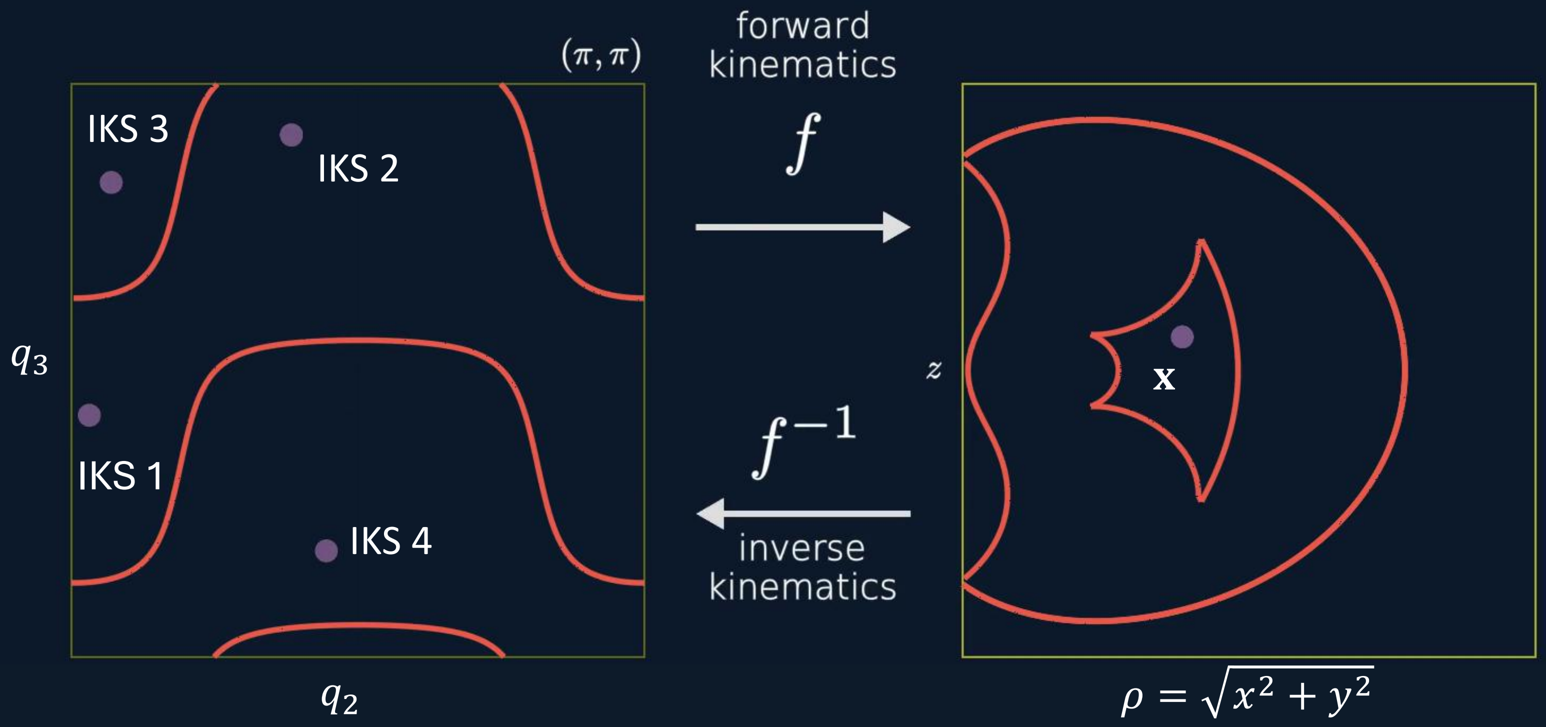
Cuspidal Robots: Robots that change inverse kinematic solutions without crossing a singularity



Joint space singularities

Workspace singularities

Research activity

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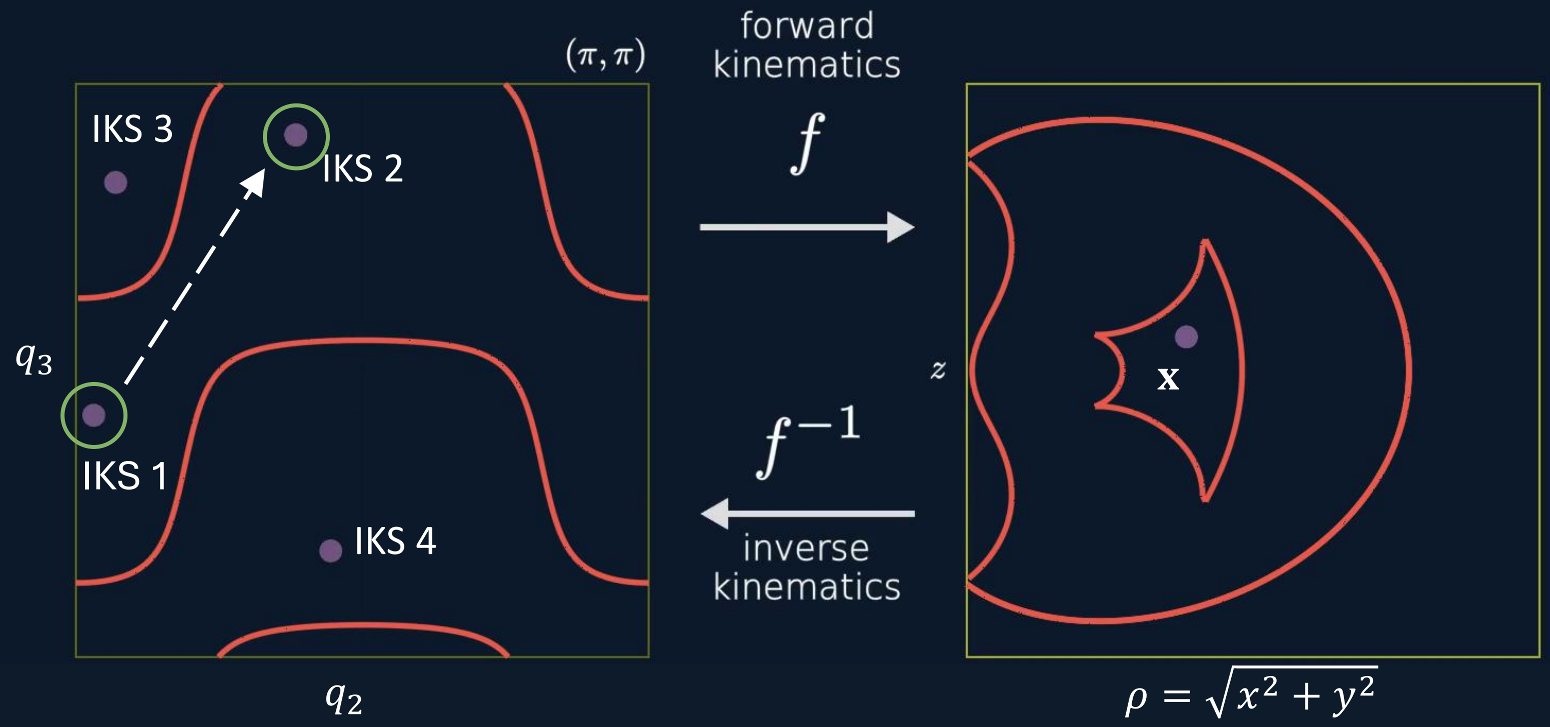


Joint space singularities

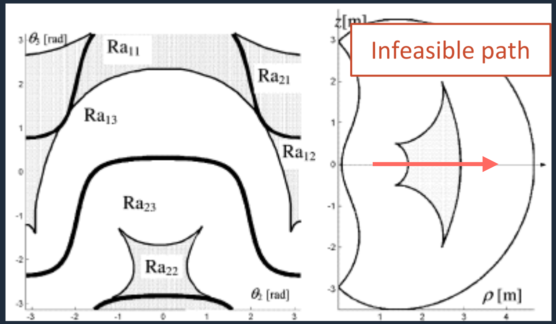
Workspace singularities

Research activity

Doctoral work

PREVIOUS WORK

- Trajectory planning issues in 3R cuspidal robots



(Wenger ‘04)

- Cuspidality analysis of 6R robots

UR5 is a noncuspidal robot

(Capco et al. ’20)

Research activity

Doctoral work

PREVIOUS WORK

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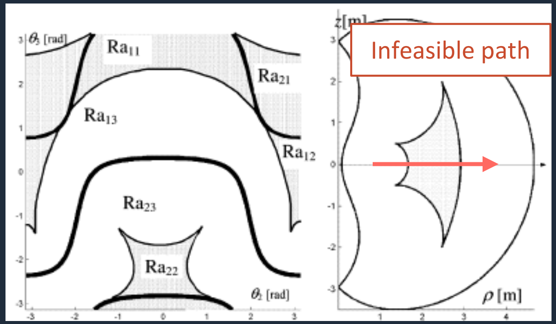

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- Cuspidality analysis of 6R robots

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1. How to determine if a given 6R robot is cuspidal or not?

2. What are the implications on path planning for a 6R cuspidal robot?

Research activity

Doctoral work

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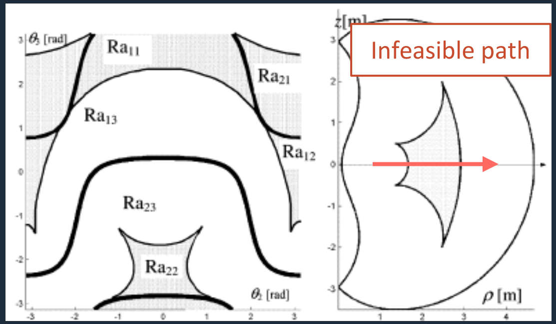

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1. How to determine if a given 6R robot is cuspidal or not?

2. What are the implications on path planning for a 6R cuspidal robot?

Certified IK + Certified connectivity

=

Deciding cuspidality

Result 1:

Even a small deviation from conventional design almost always leads to cuspidal robots.

(ISSAC'22, IJRR'24)

Conventional robot

offset

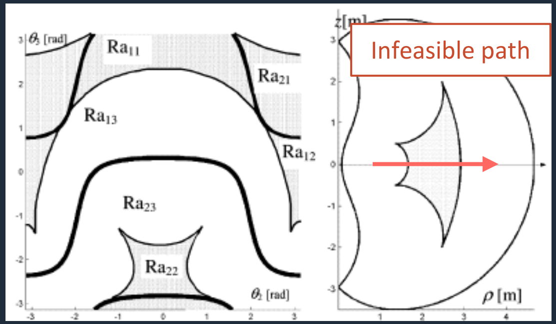
Cuspidal robot

Research activity

Doctoral work

PREVIOUS WORK

- Trajectory planning issues in 3R cuspidal robots



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- Cuspidality analysis of 6R robots

UR5 is a noncuspidal robot

(Capco et al. '20)

1. How to determine if a given 6R robot is cuspidal or not?

2. What are the implications on path planning for a 6R cuspidal robot?

Perform workspace analysis and explore different path properties for cuspidal robots.

Result 1:

Even a small deviation from conventional design almost always leads to cuspidal robots.
(ISSAC'22, IJRR'24)

Result 2:

A framework for safe and reliable path planning of commercial 6R robots.
(ICRA'23, IJRR'24)

Research activity

Postdoctoral work

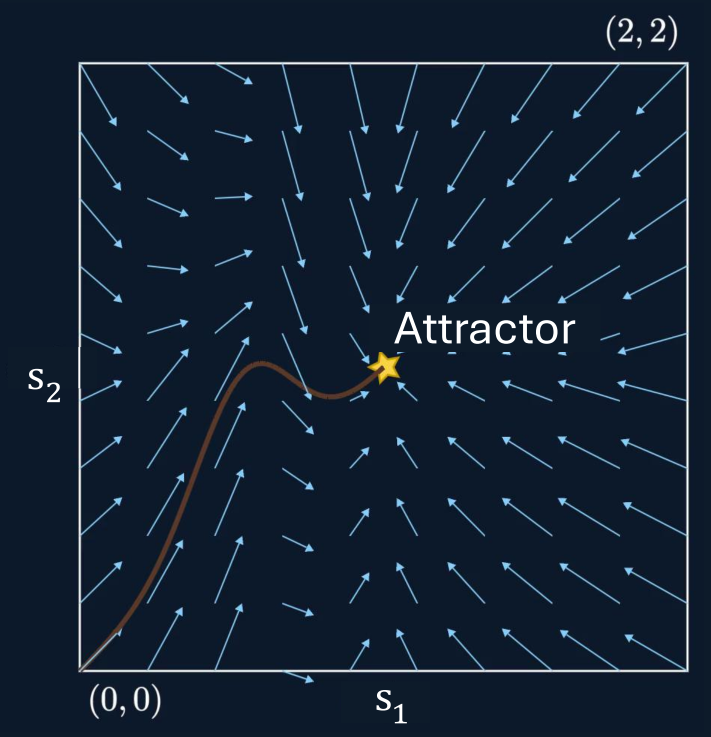
1. Explainable transfer of a user’s demonstrated behaviour across different robots.

Research activity

Postdoctoral work

Control policy: maps the robot's current state to its next action or velocity command

$$\dot{s} = f(s)$$



A control policy can be visualized as a vector field in the parameter space

1. Explainable transfer of a user's demonstrated behaviour across different robots.

Research activity

Postdoctoral work

PREVIOUS WORK

Morphology-aware transfer

1. Infer the robot morphology (Trabucco et al. ‘22)

2. Hint the learning model about robot structure with biases (Sferrazza et al., ‘24)

3. Train a single policy that handles robot variations (Mu et al, ‘26, Yang et al., ‘26)

1. Explainable transfer of a user’s demonstrated behaviour across different robots.

Research activity

Postdoctoral work

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1. Explainable transfer of a user’s demonstrated behaviour across different robots.

Embedding robot structure into learning for safe and deterministic motion.

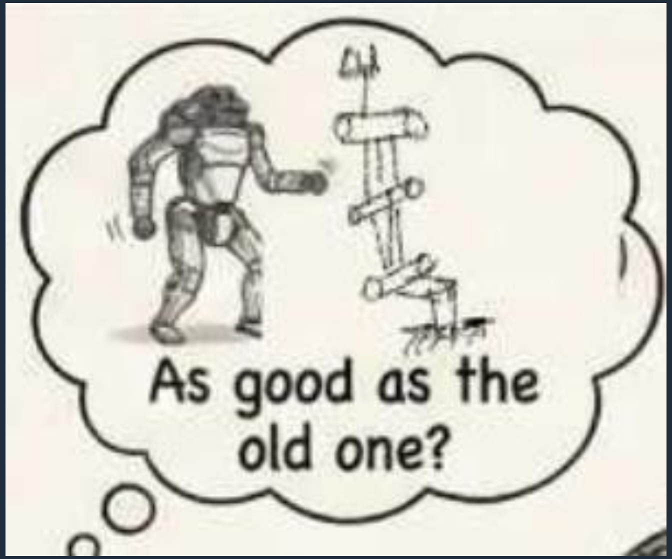
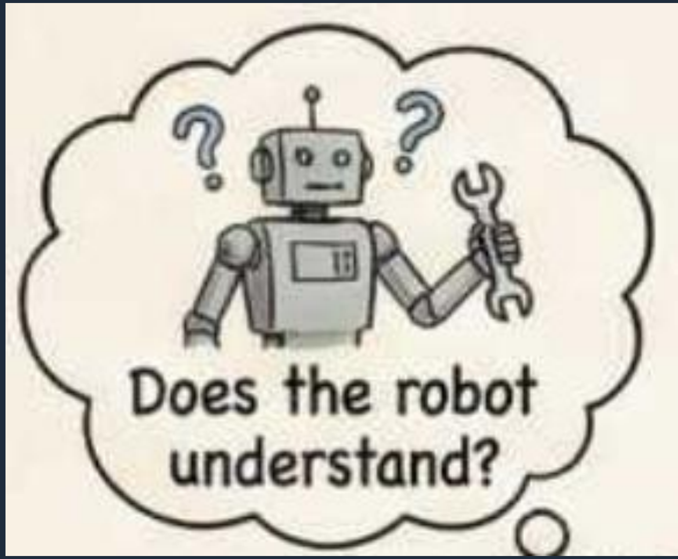
Stable robot-agnostic policy
+
Deterministic robot-specific execution
=
Explainable cross-robot skill transfer

1. Introduced a novel paradigm for an explainable cross-robot skill-transfer framework (Science Robotics ‘26)

2. Collaboration with IRI Spain on kinematic analysis using Conformal Geometric Algebra (IMA ‘25 x2)

Research plan

Scientific focus

Reliable Execution	Skill Scalability	Embodied alignment
		

Short-term

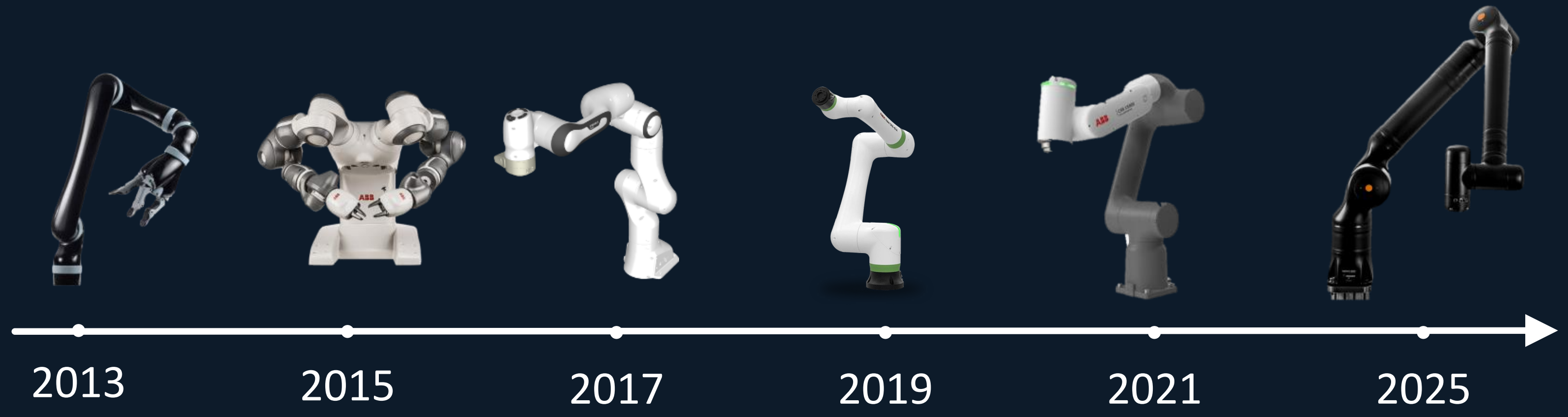
Mid-term

Long-term

Phase 1

Short-term scientific focus

Reliable Execution



1. Many modern commercial robots (6R and 7R) are cuspidal.

2. Almost all of them are used for collaborative purposes.

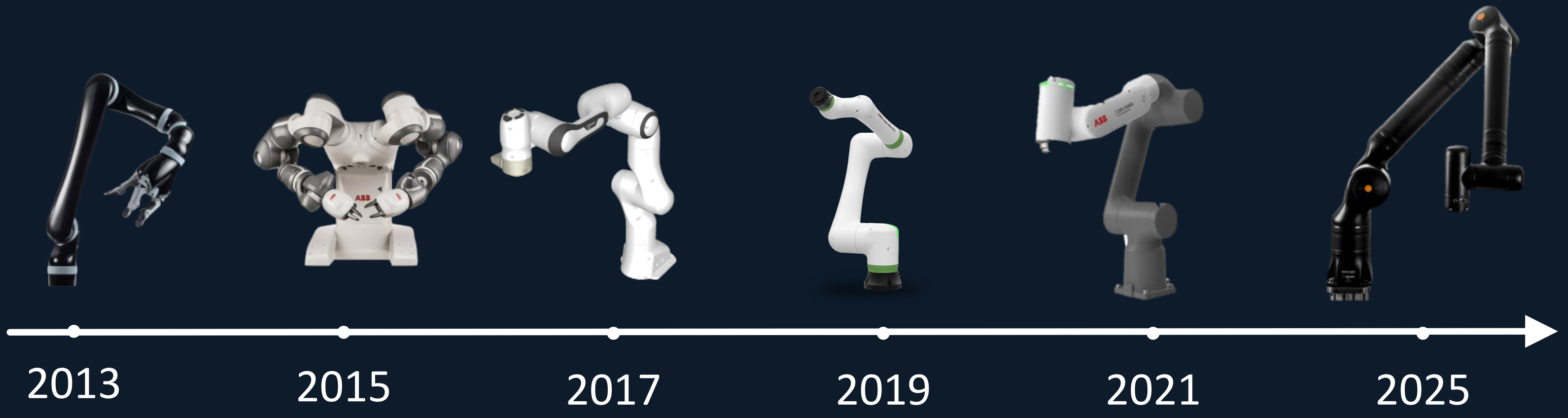
Analysis of cuspidal robots is both urgent and essential.

Phase 1

Short-term scientific focus

Reliable Execution

1. Design: Kinematic analysis of cuspidal robots suffers from the curse of dimensionality.



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Phase 1

Short-term scientific focus

Reliable Execution

1. Design: Kinematic analysis of cuspidal robots suffers from the curse of dimensionality.

Co-design approach:

1. Design: Perform global kinematic analysis for cuspidal robots.

1. Use accelerated processing and efficient algorithms for workspace analysis

Efficient IK solvers	+	Reduced-dimension analysis	+	Certified cell decomposition	=	Global kinematics of cuspidal robots
Conformal Geometric Algebra		Redundant angle parameterization		Cylindrical Algebraic Decomposition		

Phase 1

Short-term scientific focus

Reliable Execution

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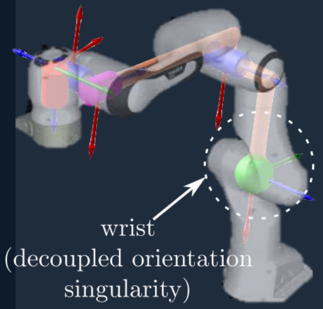
Conformal Geometric Algebra

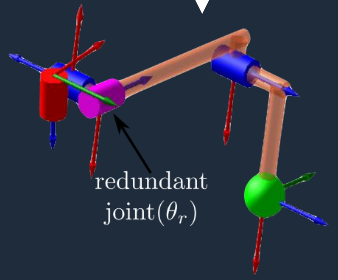
Redundant angle parameterization

example

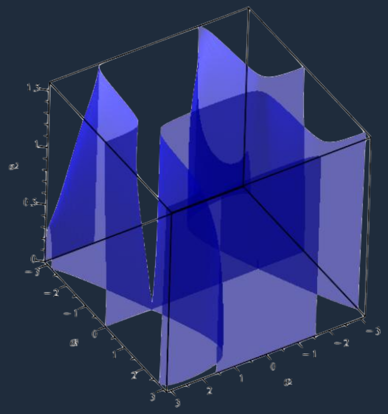
Cylindrical Algebraic Decomposition



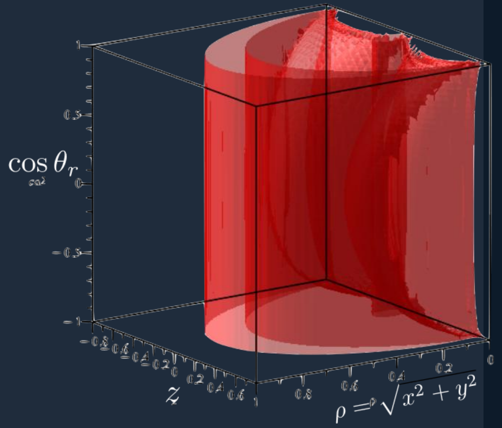
7R robot



4R + 3R robot



Stacked joint space singularities



Stacked workspace singularities

Phase 1

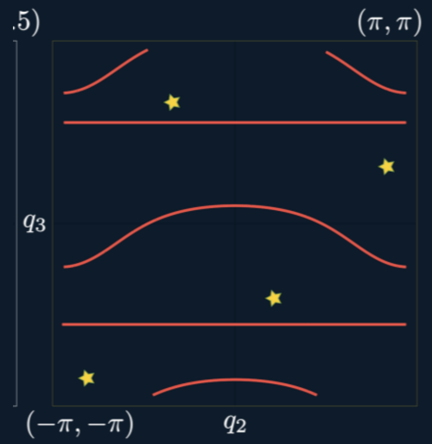
Short-term scientific focus

Reliable Execution

1. Design: Kinematic analysis of cuspidal robots suffers from the curse of dimensionality.

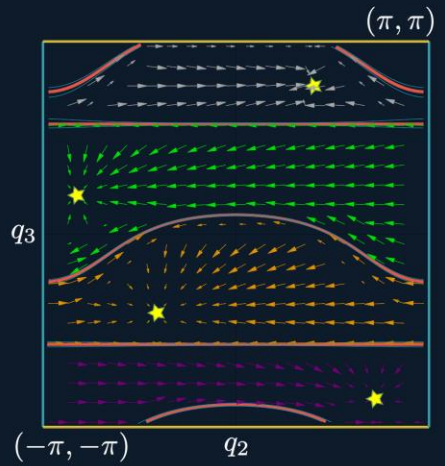
2. Control: No unique isolated control policies for cuspidal robots.

Noncuspidal robot



Joint space
'star' is IKS of attractor

One IKS → One control policy



Joint space policy
Individual policy in each region

Phase 1

Short-term scientific focus

Reliable Execution

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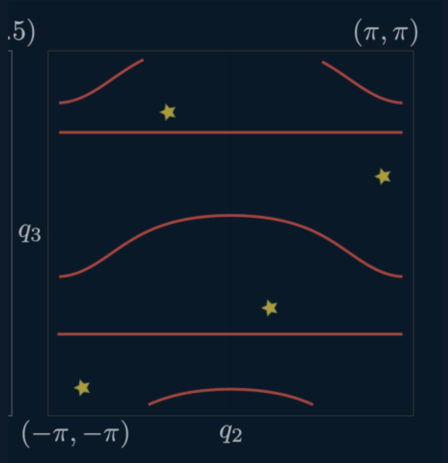
Noncuspidal robot



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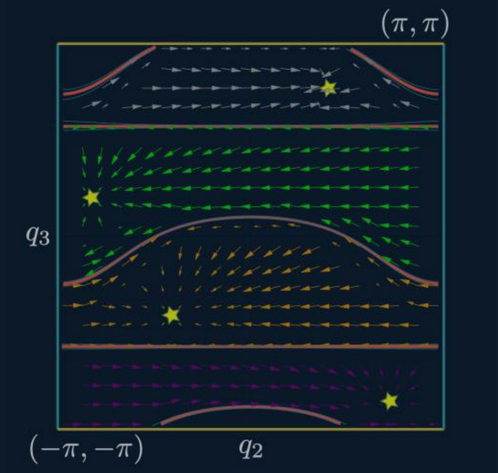
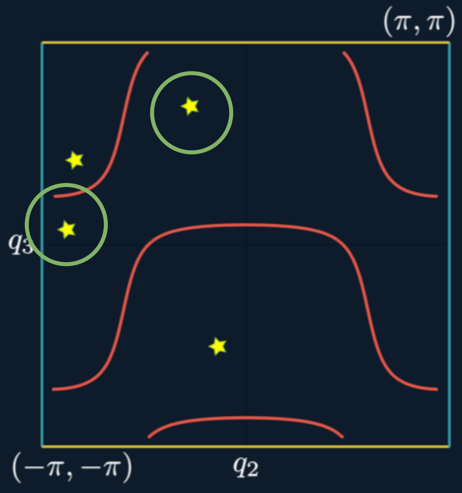


Joint space policy
Individual policy in each region

Cuspidal robot



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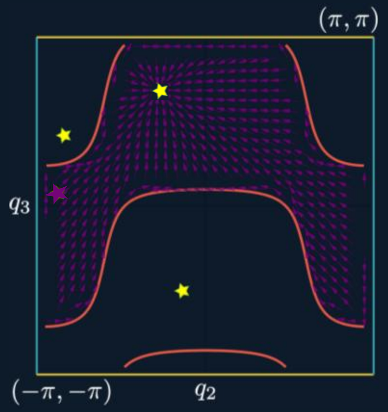
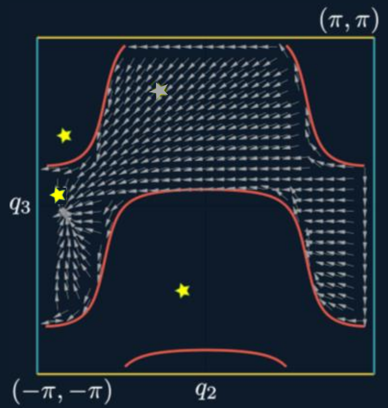




Which control policy to use?

Phase 1

Short-term scientific focus

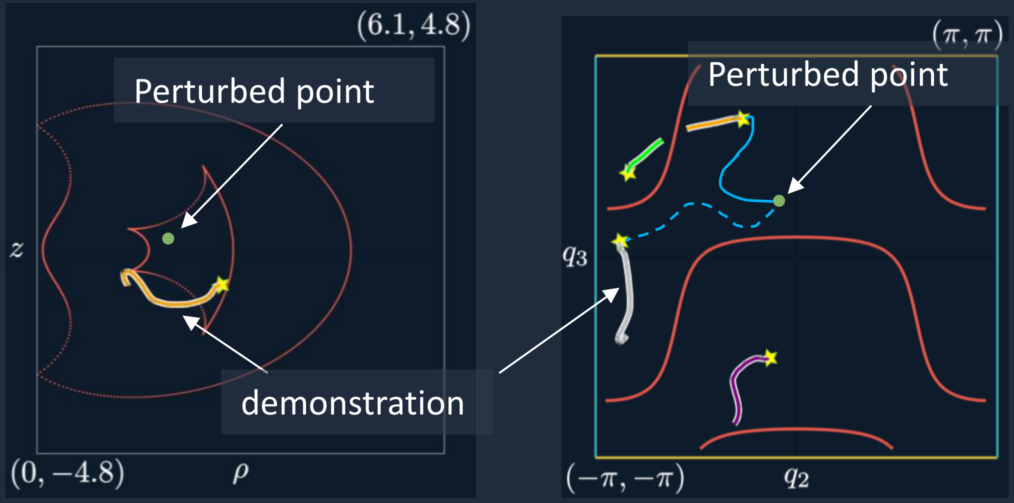
Reliable Execution

1. Design: Kinematic analysis of cuspidal robots suffers from the curse of dimensionality.
2. Control: No unique isolated control policies for cuspidal robots.

Co-design approach:

1. Design: Perform global kinematic analysis for cuspidal robots.
2. Control: Develop metrics to identify the suitable control policy

2. Study a metric to enquire proximity to user intent



workspace

joint space

Which path do I choose?

Leverage Kinematic Intelligence

Phase 1

Short-term scientific focus

Reliable Execution

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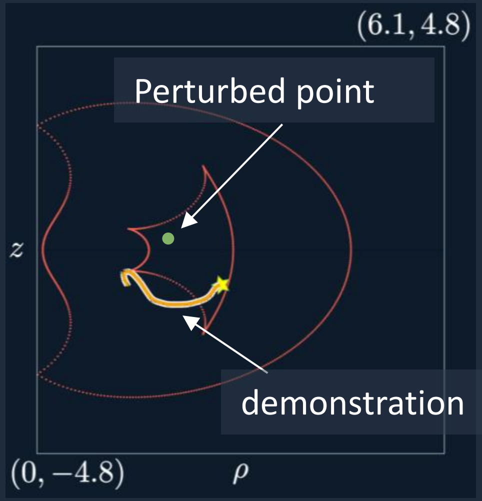
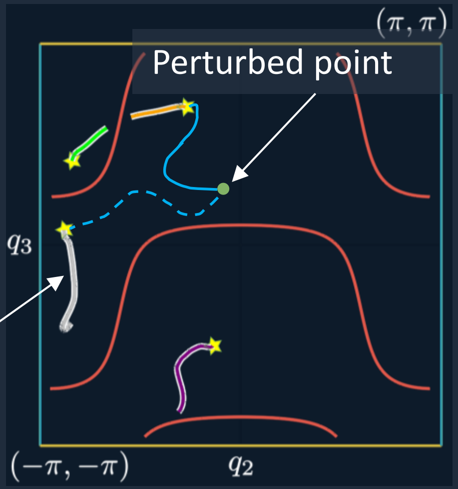
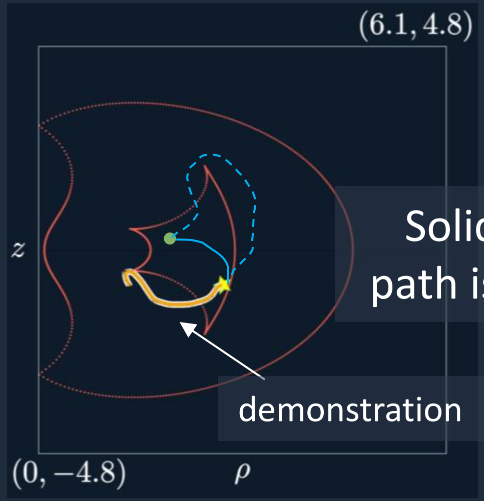




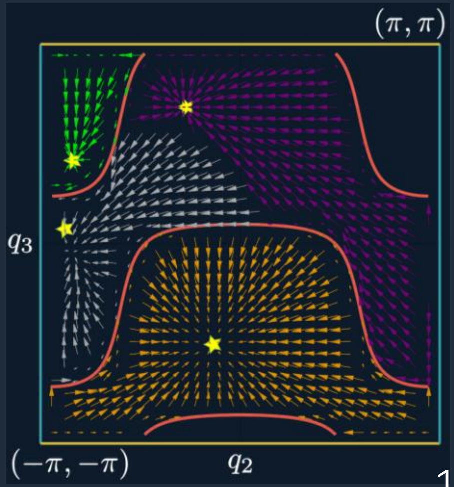
Which path do I choose?

Use fast all-at-once forward mapping and measure the deviation from the intended behaviour



Solid blue path is better

Expected final control policy

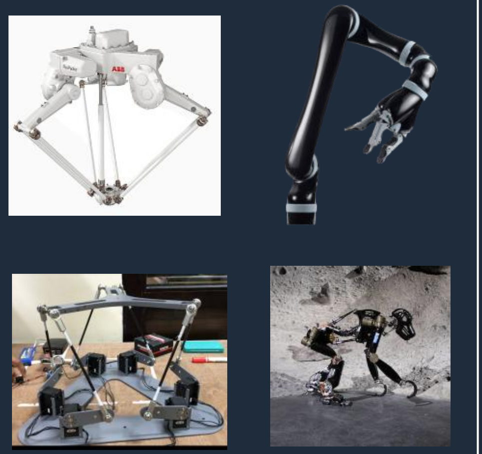


Phase 2

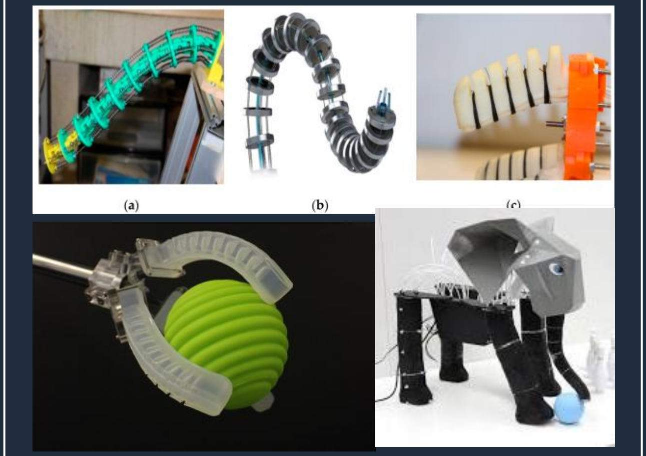
Mid-term scientific focus

Skill Scalability

Rigid links



Soft material



Increasing flexibility

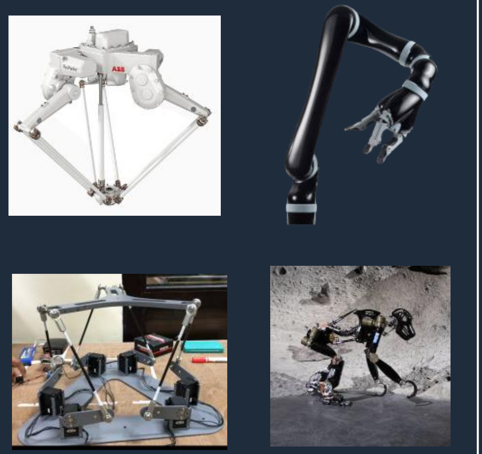
Increasing precision

Phase 2

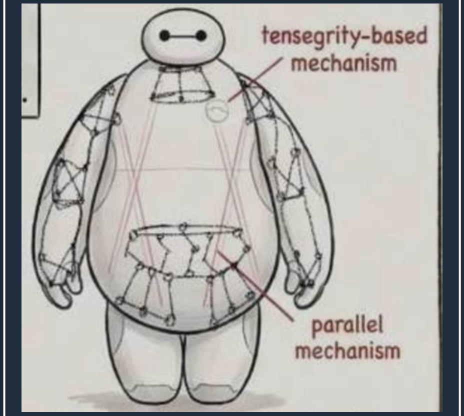
Mid-term scientific focus

Skill Scalability

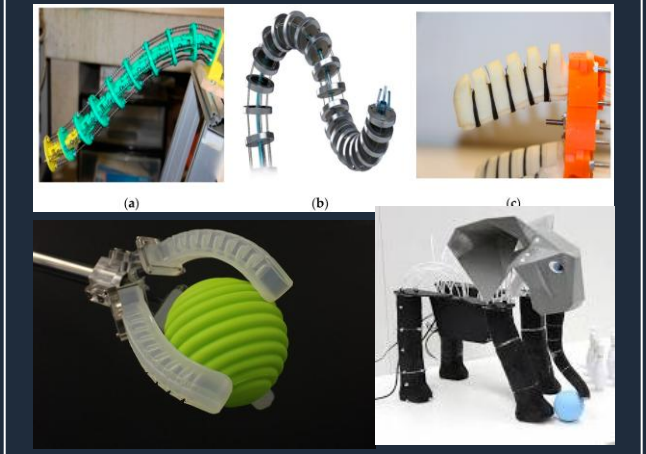
Rigid links



Robots for HRI



Soft material



Increasing flexibility

Increasing precision

Properties desired in human-robot interaction (Haddadin and Croft, ‘16):

1. Mechanical impedance
2. Lightweight designs
3. Transparency (natural motion)

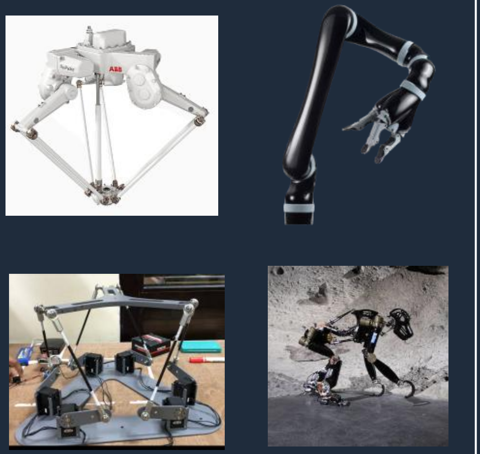
Phase 2

Mid-term scientific focus

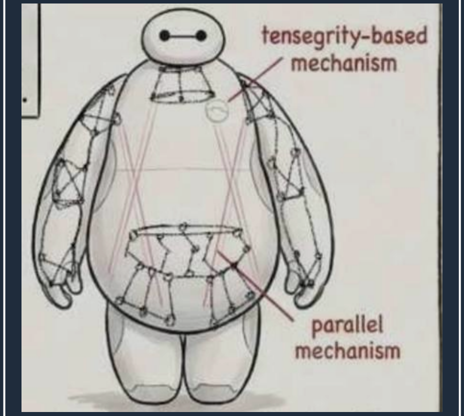
Skill Scalability

1. Design: Kinematic analysis of novel joints and hybrid mechanisms is missing.

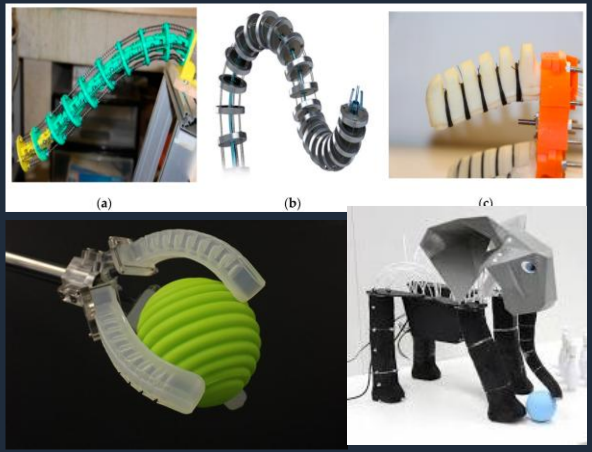
Rigid links



Robots for HRI



Soft material

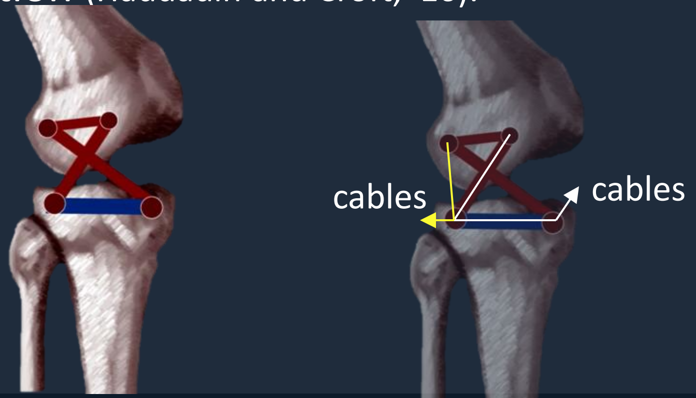


Increasing flexibility

Increasing precision

Properties desired in human-robot interaction (Haddadin and Croft, '16):

1. Mechanical impedance
2. Lightweight designs
3. Transparency (natural motion)



Phase 2

Mid-term scientific focus

Skill Scalability

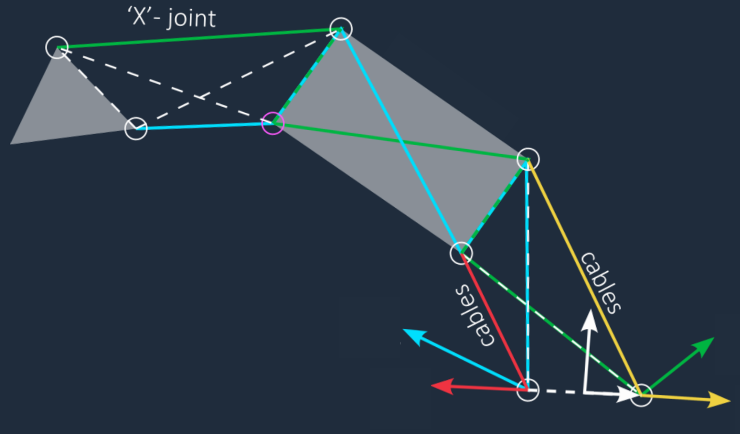
1. Design: Kinematic analysis of novel joints and hybrid mechanisms is missing.

Co-design approach:

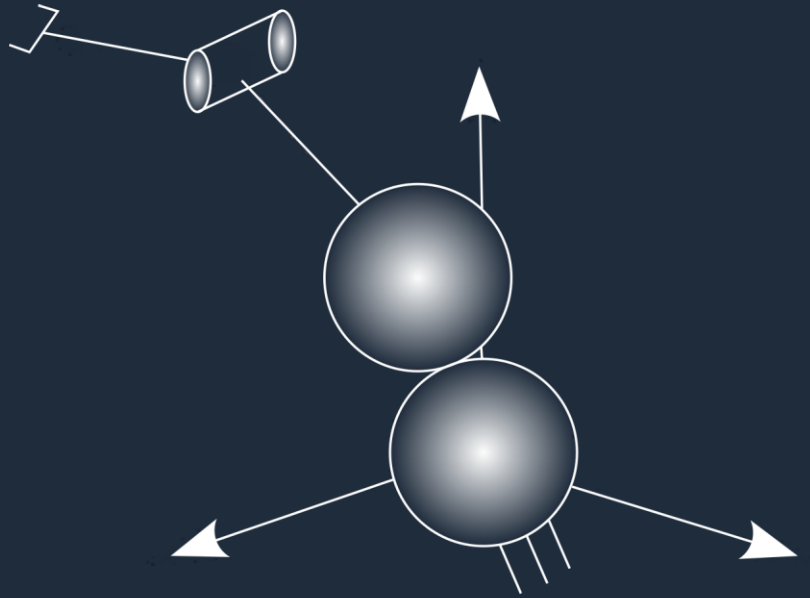
1. Design: Rigid–soft trade-off through novel joints and mechanisms

1. Rigid–soft trade-off through novel joints and mechanisms

Mechanisms that mimic human joints + Cable-driven actuation = Light robots with variable stiffness



Cable-driven 'X'-joints
(Muralidharan et al., '25)



Series-parallel arm with sphere-on-sphere rolling kinematics

Explore hybrid designs and novel actuations to maximize the desirable properties for HRI

Phase 2

Mid-term scientific focus

Skill Scalability

1. Design: Kinematic analysis of novel joints and hybrid mechanisms is missing.

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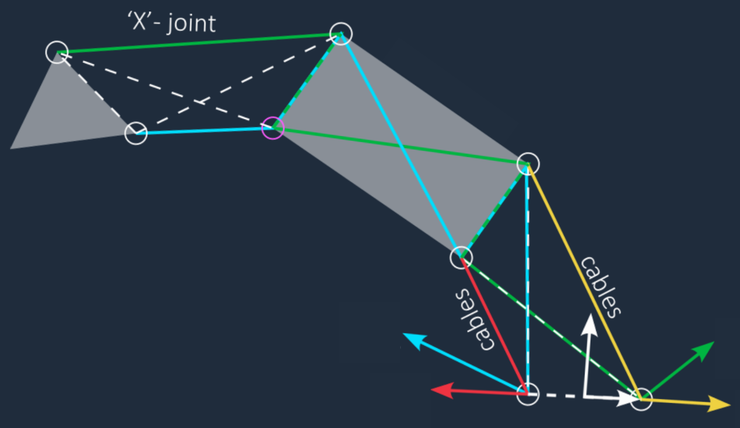

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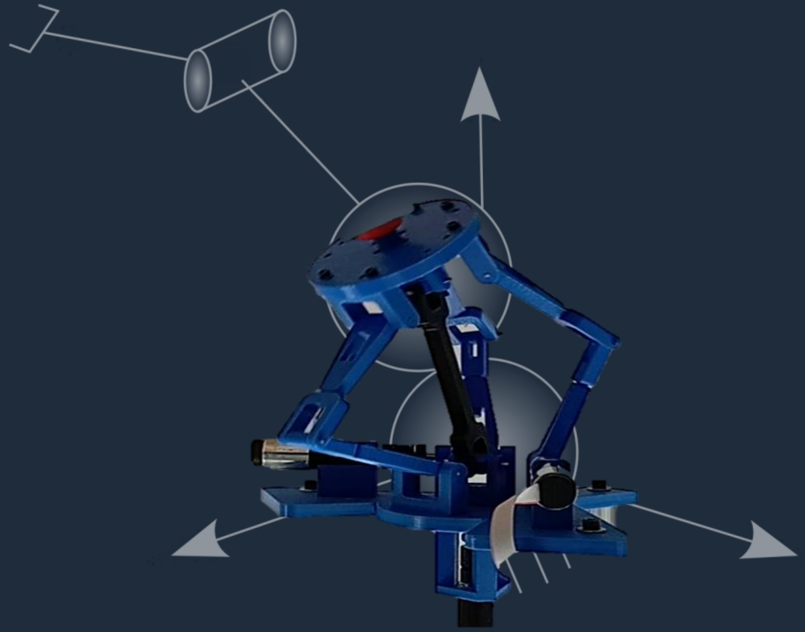


Series-parallel arm with sphere-on-sphere rolling kinematics

Explore hybrid designs and novel actuations to maximize the desirable properties for HRI

Phase 2

Mid-term scientific focus

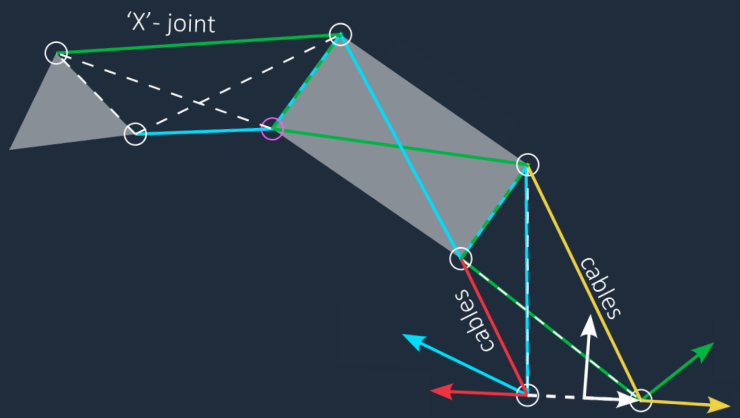
Skill Scalability

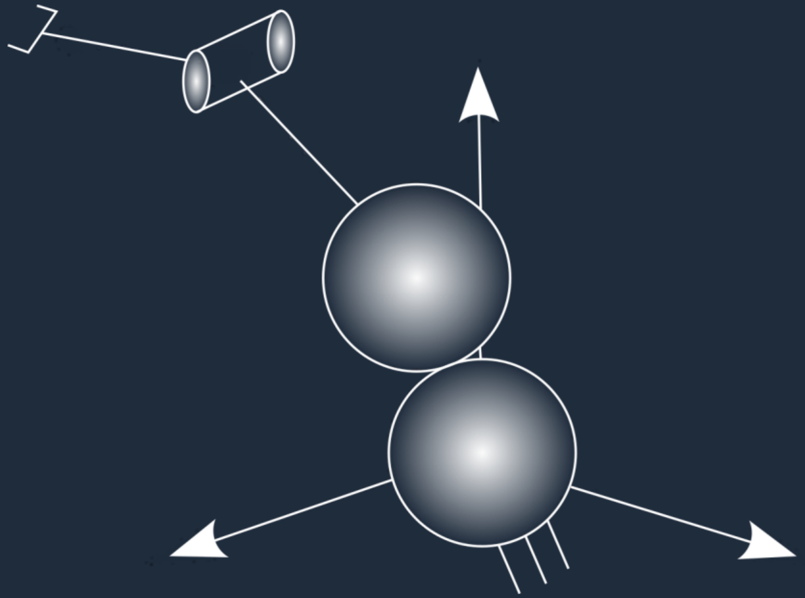
1. Design: Kinematic analysis of novel joints and hybrid mechanisms is missing.

2. Control: Skill transfer framework that works across present robots and future designs.

2. Explainable skill transfer through Kinematic Intelligence

Older generation robots





Novel mechanisms

Phase 2

Mid-term scientific focus

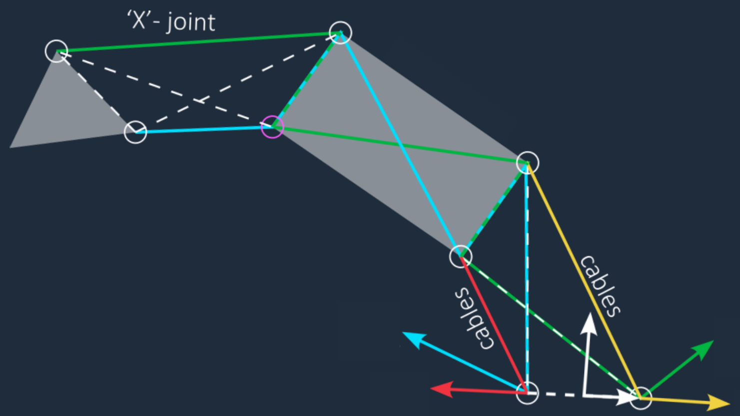
Skill Scalability

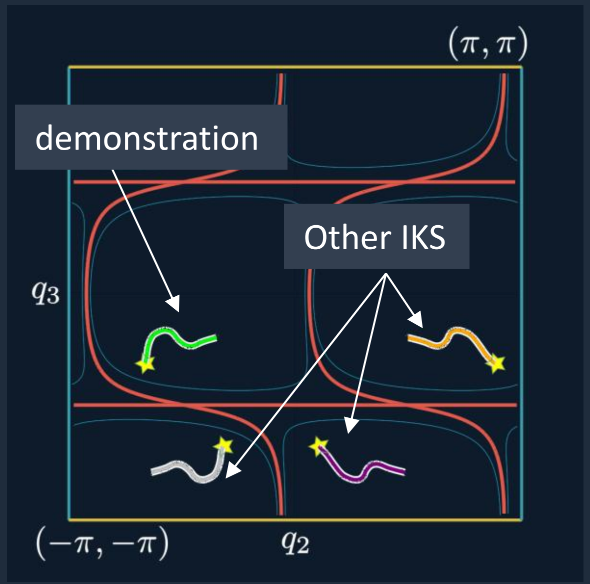
1. Design: Kinematic analysis of novel joints and hybrid mechanisms is missing.
2. Control: Skill transfer framework that works across present robots and future designs.

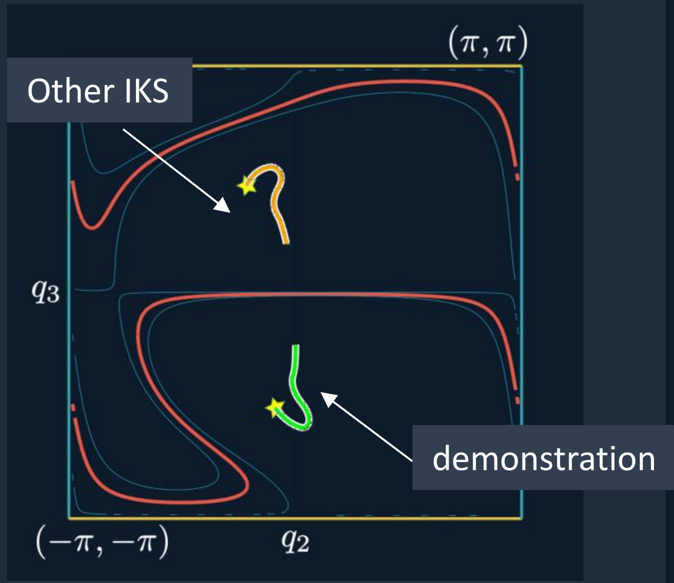
Co-design approach:

1. Design: Rigid–soft trade-off through novel joints and mechanisms
2. Control: Explainable skill transfer through Kinematic Intelligence

2. Explainable skill transfer through Kinematic Intelligence







Example singularities and IKS distribution that characterizes a robot

Phase 2

Mid-term scientific focus

Skill Scalability

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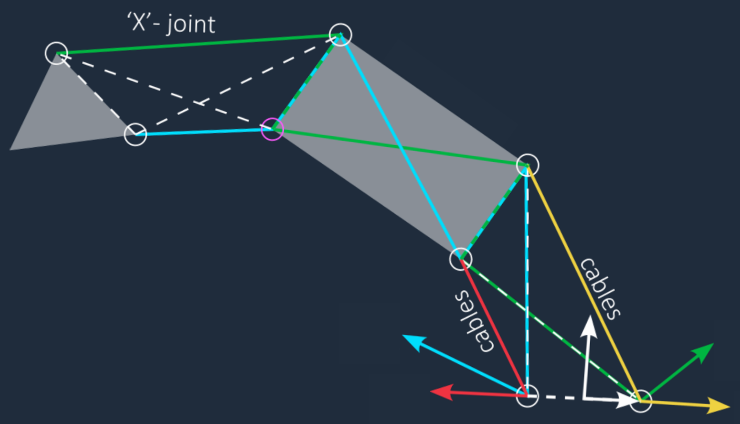
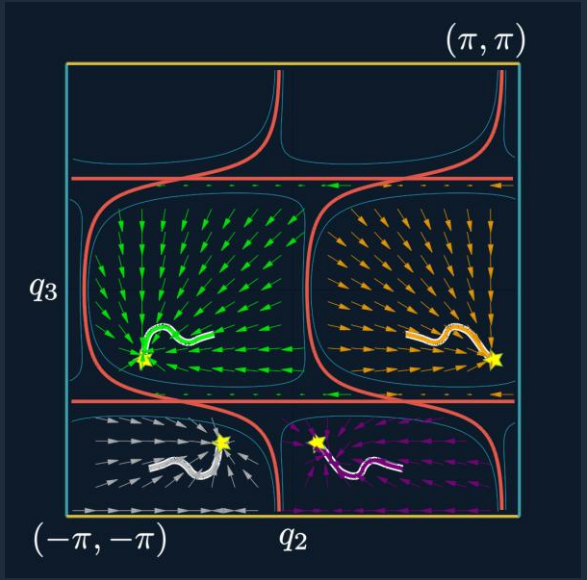
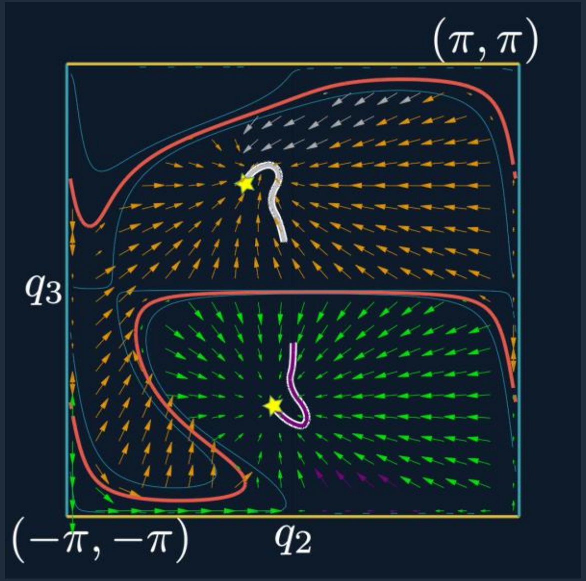






Example resulting control policies post-transfer

Integration

Choice of host laboratories

Team RDH

Le laboratoire des sciences de l'ingénieur, de l'informatique et de l'imagerie (UMR7357) (ICube)

Strasbourg

Team Gepetto

Laboratoire de recherche spécialisé dans l’analyse et l’architecture des systèmes (UPR 8001) (LAAS)

Toulouse

Integration

Team RDH, ICube, Strasbourg

Parallel robots	Medical robotics	Human-robot interaction	Complements team's existing expertise
Pierre Renaud	Florent Nageotte Benoit Rosa	**Hassan Omran**	
Kinematic intelligence	Path Plannning	Design Optimisation	I bring my research expertise
Geometric Algebra	Learning from demonstration	Transfer learning	I contribute in expanding the current expertise of the team
Olivier Piccin		Nicolas Padoy	

Integration

Team RDH, ICube, Strasbourg

1. Collaborative robots, redundant robots (x 2) (Fig. A)

2. Haptic teleoperating system (Fig: B)

3. Strong collaboration with medical teams (IHU + Camma) (Fig. B, C)

4. Project **IMARA:** develop a lightweight robotic assistant for the operating room.

Figure A

Figure B

Figure C

Integration

Team RDH, ICube, Strasbourg

1. Collaborative robots, redundant robots (x 2) (Fig. A)
2. Haptic teleoperating system (Fig: B)
3. Strong collaboration with medical teams (IHU + Camma) (Fig. B, C)
4. Project **IMARA:** develop a lightweight robotic assistant for the operating room.

Perfect environment to extend my research in medical applications

Figure A

Figure B

Figure C

Integration

Team Gepetto, LAAS, Toulouse

Path planning

Optimisation

Transfer learning

Complements team's existing expertise

Philippe Souères
Florent Lamiraux

Nicolas Mansard

Olivier Stasse

Kinematic Intelligence

Cuspidal robots

Parallel robots

I bring my research expertise

Co-design

Learning from demonstration

I contribute in expanding the current expertise of the team

Co-design approach

Reliable Execution

Phase 1: Year 0-2

Cuspidal robots' singularities and control policy ambiguity

Use modern tools to study the workspace and control policy

Skill Scalability

Phase 2: Year 2-5

Skill transfer among current and future generations of robots

Kinematic Intelligence for skill transfer in current & future robots

Embodied Alignment

Phase 3: Year 5+

Learning based interactions need to be analytically grounded

Explore coupling of motion with language-based task planning

Publications

- 8 journal articles: Science Robotics, IJRR, IEEE RA-L, MMT (x2), RCIM, JMD, SI
- 13 international conferences: ICRA, IROS, ARK, ISSAC, EUCOMES, CK, IMA (x2), RAHA, CASE, CFM, RAAD
- 2 open-source softwares: ParaOpt, Skill Transfer (Apr, '26)

Communications

- 6 Invited talks: SIMERO'23, RICAM'24, EPFL, Strasbourg, Toulouse, Bengaluru
- 3 thematic events: Swiss Robotics Day (x2), RSS'26 Workshop
- 1 Media article: EPFL News (April, 2026)

Teaching

- 20 hrs CM: Introduction to Robotics, Ecole Centrale de Nantes
- 20 hrs CM: Kinematically grounded motion planning, Ecole Polytechnique Federale de Lausanne
- 96 hrs TP + TD: Design of robots, Collaborative robotics, Optimal kinematic design of robots

References

Cheney et al., 2018 — Cheney, N., Bongard, J., SunSpiral, V., and Lipson, H., Scalable co-optimization of morphology and control in embodied machines. Journal of Royal Society Interface.

Wenger et El Omri, 1996 — P. Wenger and J. El Omri, "Changing posture for cuspidal robot manipulators," *Proceedings of IEEE ICRA*, Minneapolis, MN, USA, 1996

Wenger, 2004 — P. Wenger, Uniqueness Domains and Regions of Feasible Paths for Cuspidal Manipulators, Transaction on Robotics, 2004

Capco et al., 2020 — Capco. J, Safey El Din. M, and Schicho. J, Robots, computer algebra and eight connected components, International Symposium on Symbolic and Algebraic Computation, 2020

Trabucco et al., 2022 — Brandon Trabucco, Mariano Phielipp, Glen Berseth *Proceedings of the 39th International Conference on Machine Learning*, PMLR 162:21677-21691, 2022.

Sferazza et al., 2024 — Carmelo Sferrazza and Dun-Ming Huang and Fangchen Liu and Jongmin Lee and Pieter Abbeel, Body Transformer: Leveraging Robot Embodiment for Policy Learning, arXiv, 2024

Mu et al., 2026 — Juncheng Mu and Sizhe Yang and Hojin Bae and Feiyu Jia and Qingwei Ben and Boyi Li and Huazhe Xu and Jiangmiao Pang, One-Policy-Fits-All: Geometry-Aware Action Latents for Cross-Embodiment Manipulation, arXiv, 2026

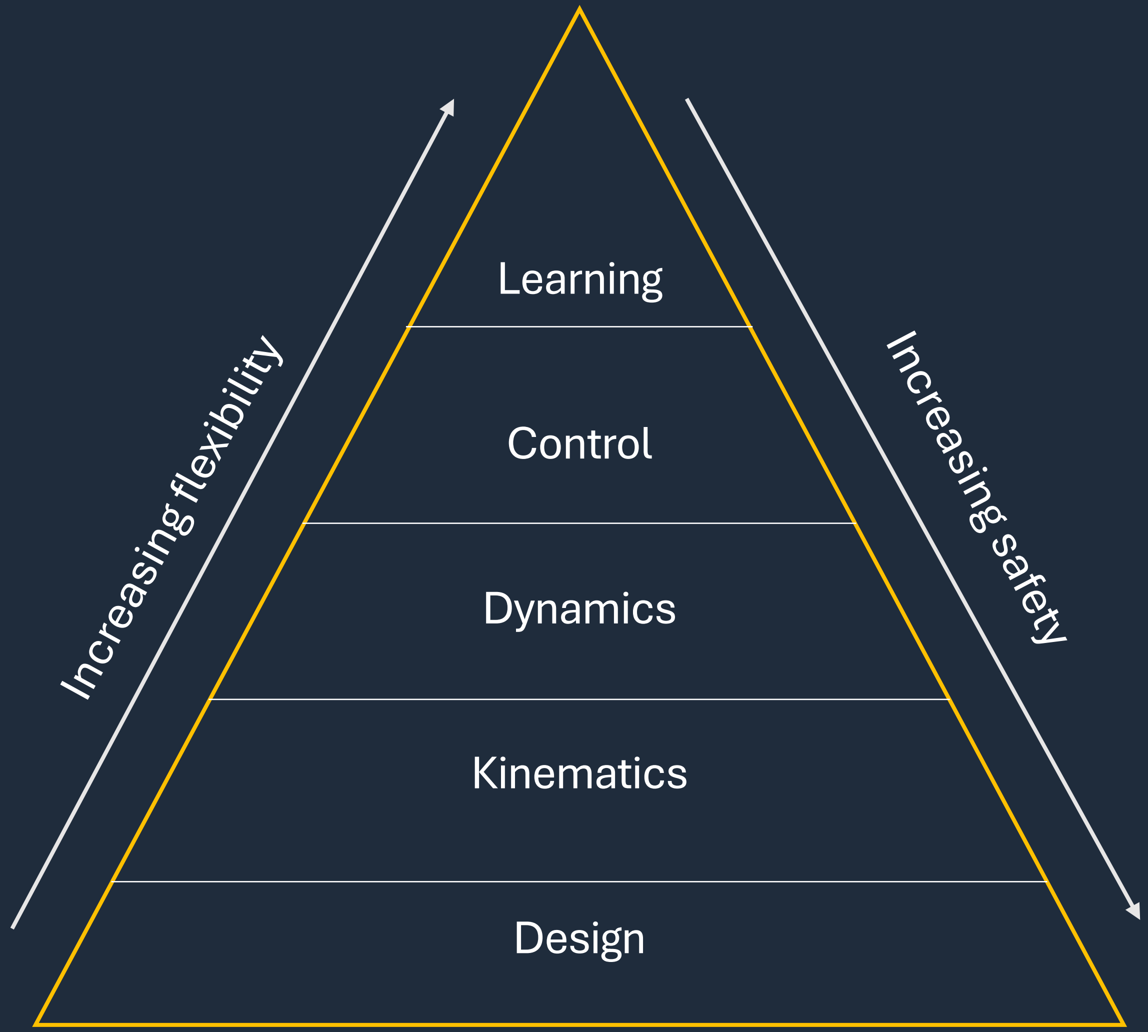
Yang et al., 2026 — Jonathan Yang and Chelsea Finn and Dorsa Sadigh, Data Analogies Enable Efficient Cross-Embodiment Transfer, arXiv, 2026

Phase 3

Long-term scientific focus

Grounded Interaction

1. Deterministic embedding of language with motion is essential
2. Designs developed using generative models need kinematic validation



Phase 3

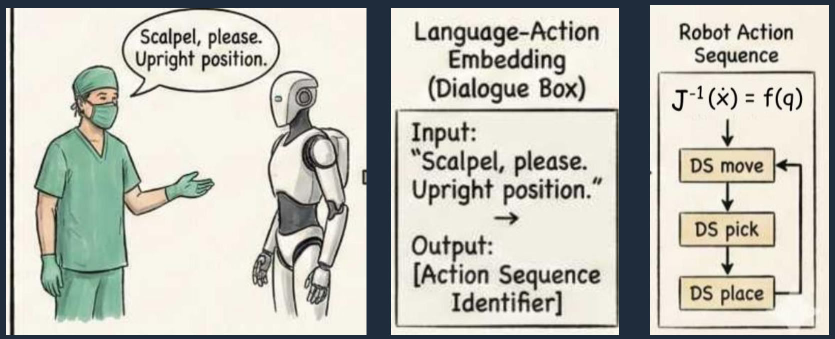
Long-term scientific focus

Grounded Interaction

1. Deterministic embedding of language with motion is essential
2. Designs developed using generative models need kinematic validation

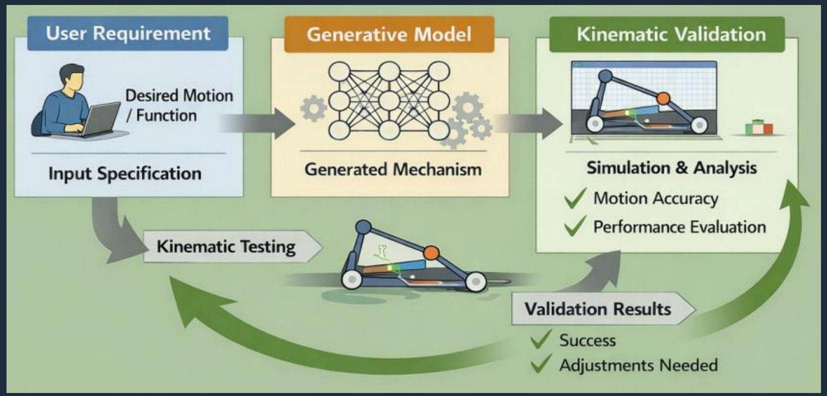
Co-design approach:

1. Develop motion language framework by mapping language tokens with DS
2. Develop generative models that are aware of kinematic analysis.



Language-based safe and explainable HRI

- Investigate the coupling of language and motion primitives
- Leverage task planning from language models, and guarantee motion with Kinematic intelligence

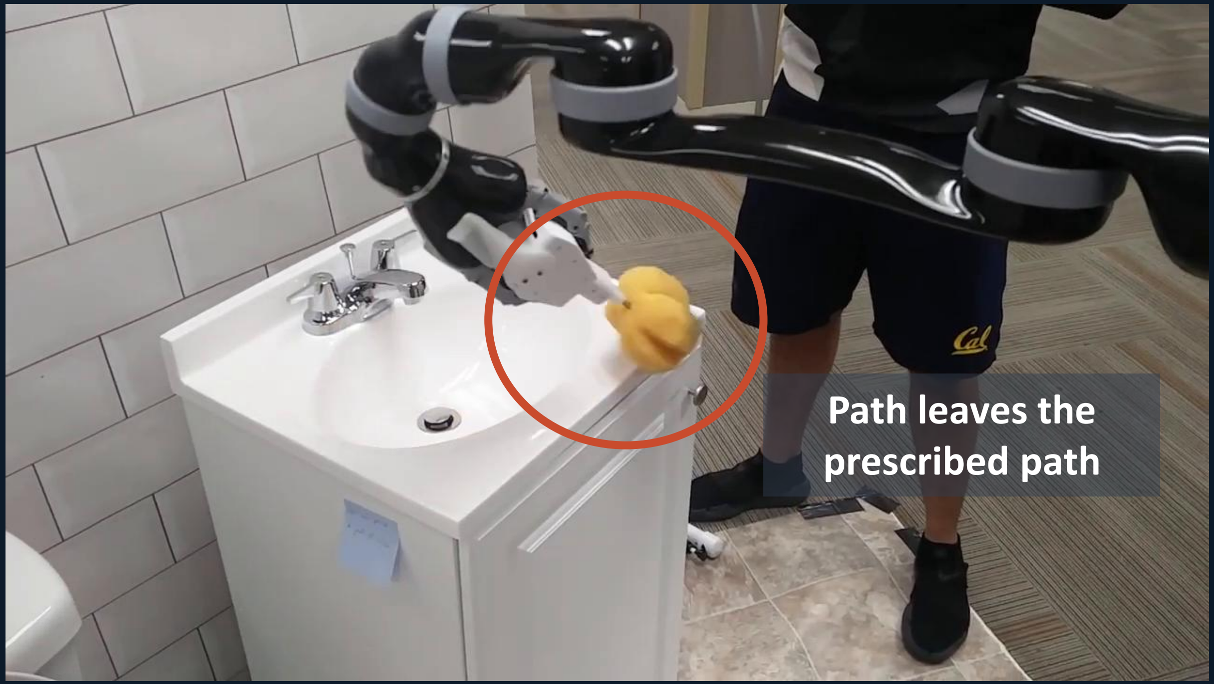


Kinematic validation of generative models

- Stage 1: Use a separate kinematic validation pipeline to verify designs from the generative model
- Stage 2: couple the kinematic understanding of the mechanisms in the model itself

Research activity

Doctoral work



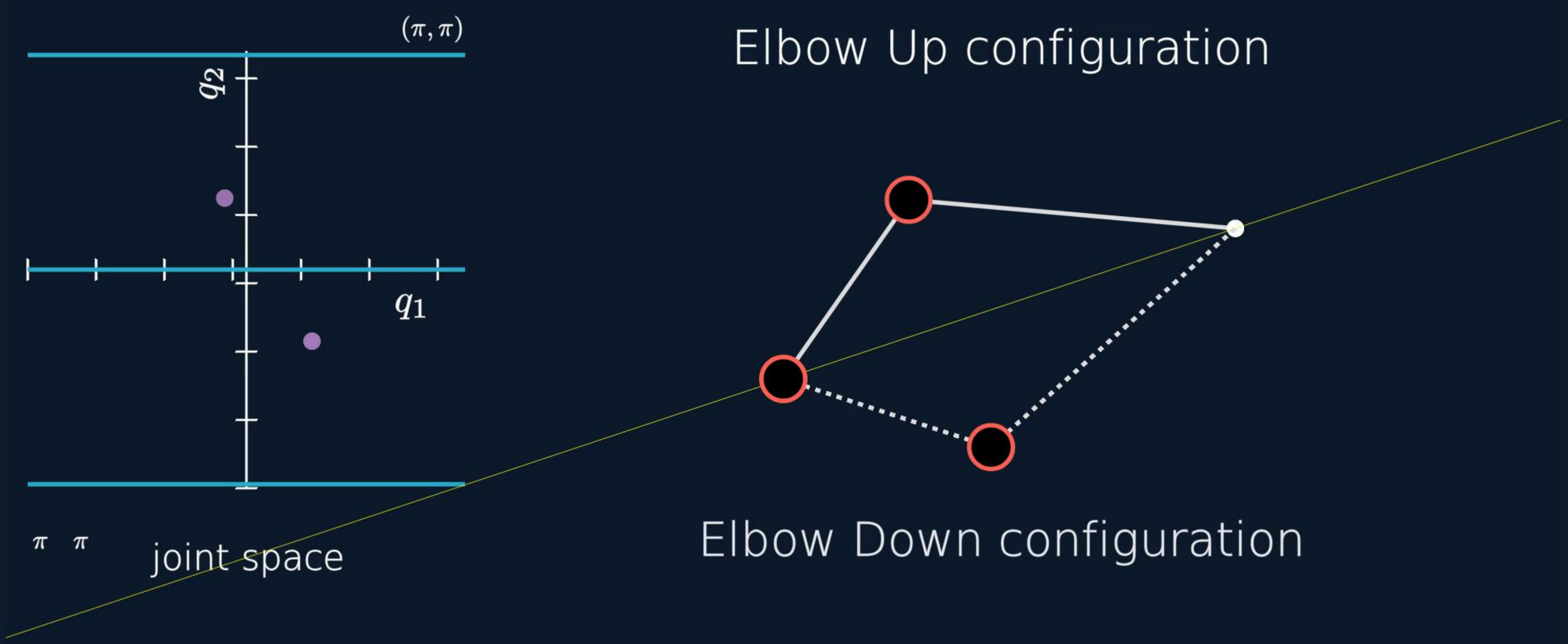
Catastrophic failure

Path planning failure of cuspidal robots in real world application

Source: Achille Verheye (achille0.medium.com)

Research activity

Doctoral work

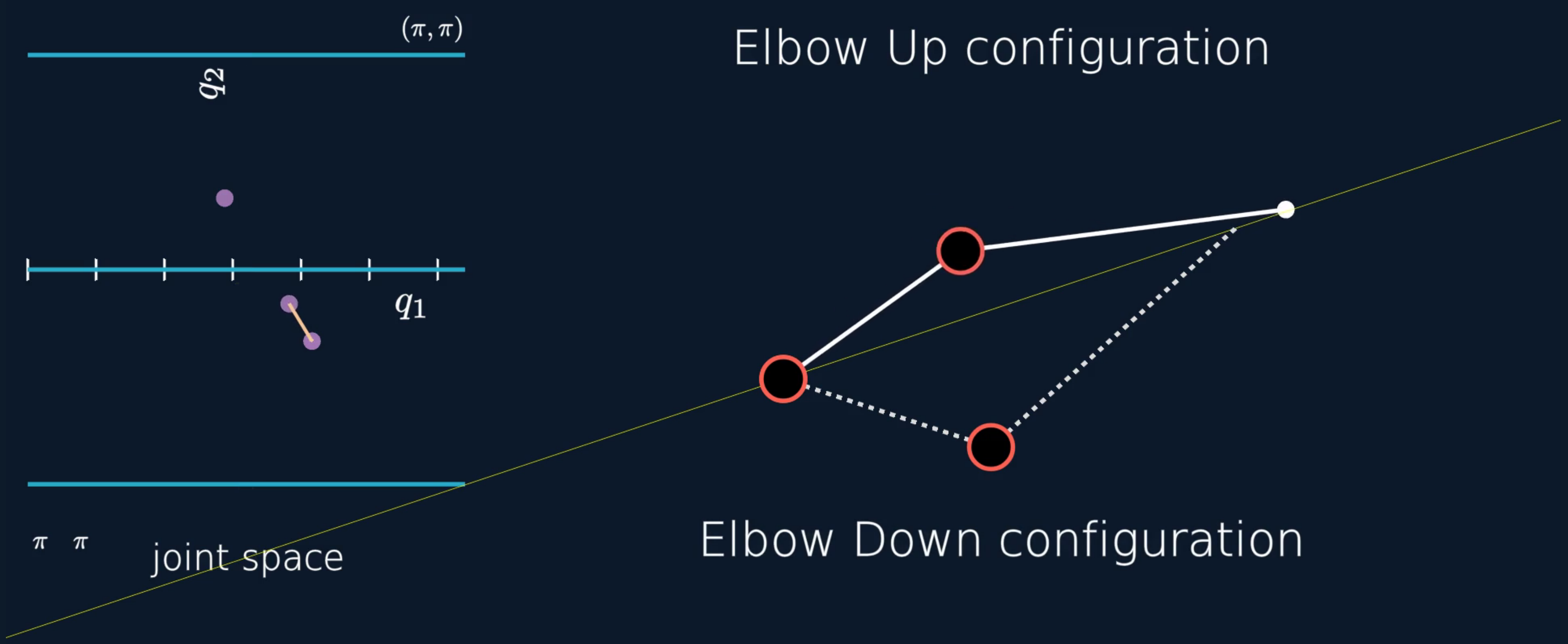


Inverse kinematic solutions are separated by singularities, providing a clear distinction of the paths

Noncuspidal robot

Research activity

Doctoral work



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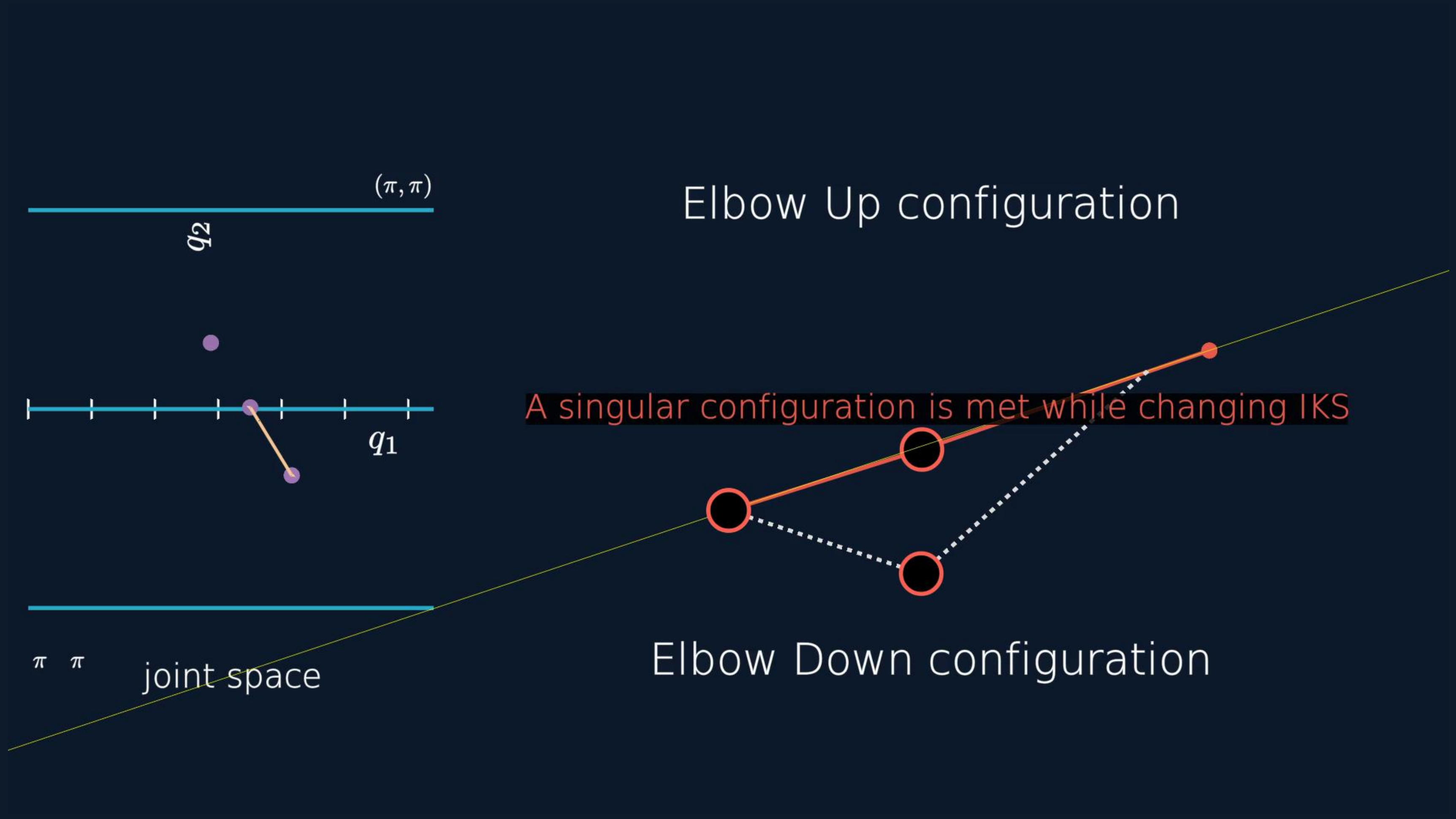
Noncuspidal robot

Research activity

Doctoral work

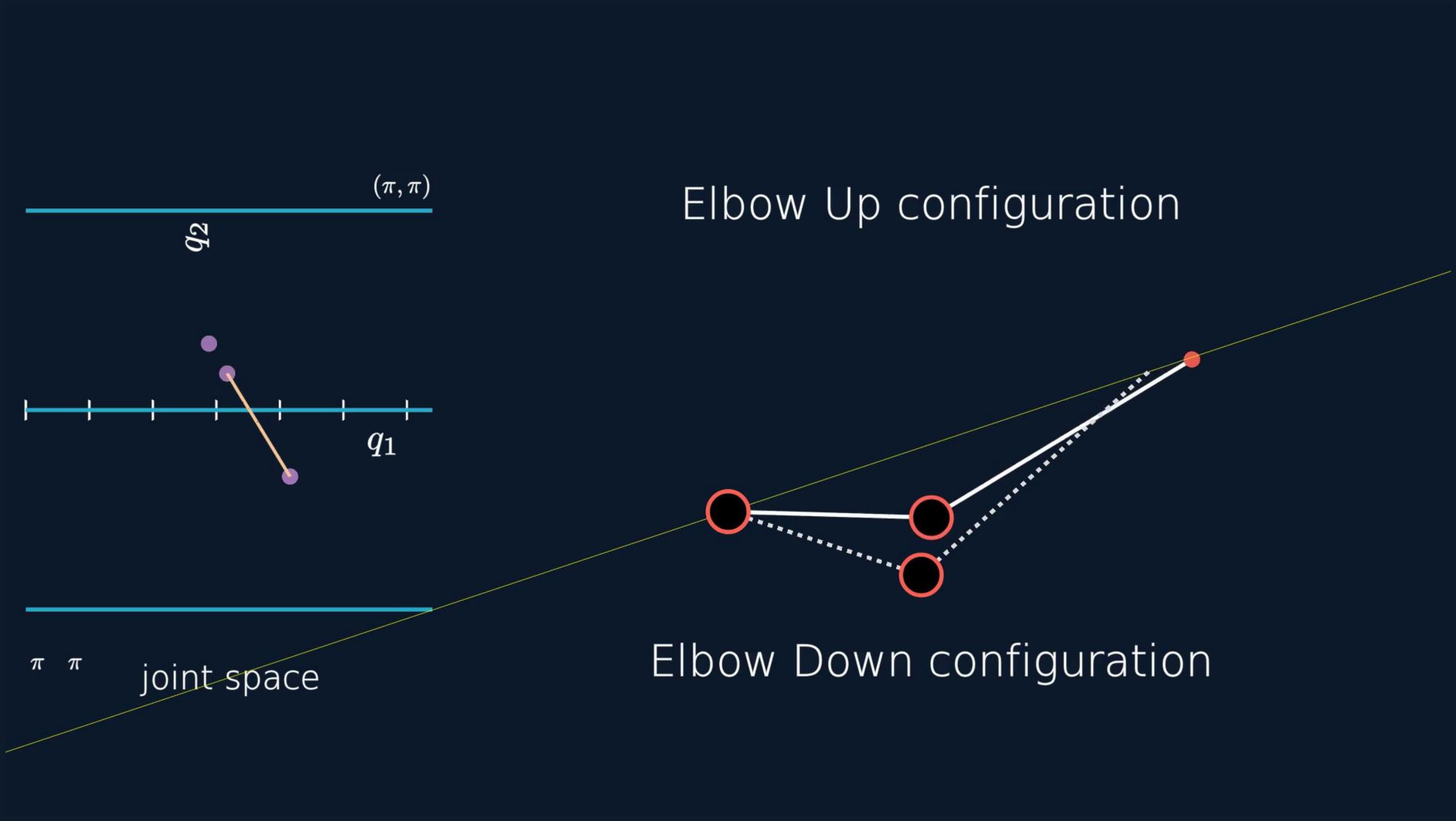


Noncuspidal robot

Inverse kinematic solutions are separated by singularities, providing a clear distinction of the paths

Research activity

Doctoral work

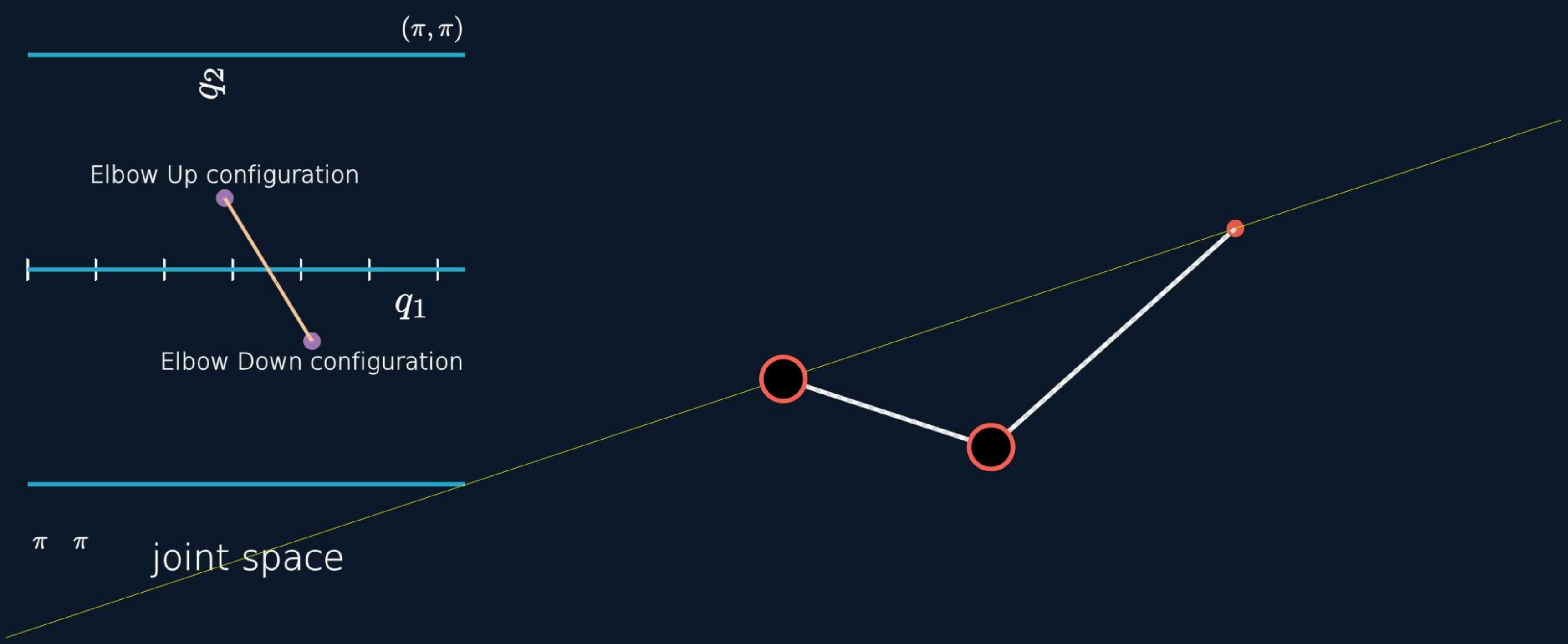


Inverse kinematic solutions are separated by singularities, providing a clear distinction of the paths

Noncuspidal robot

Research activity

Doctoral work

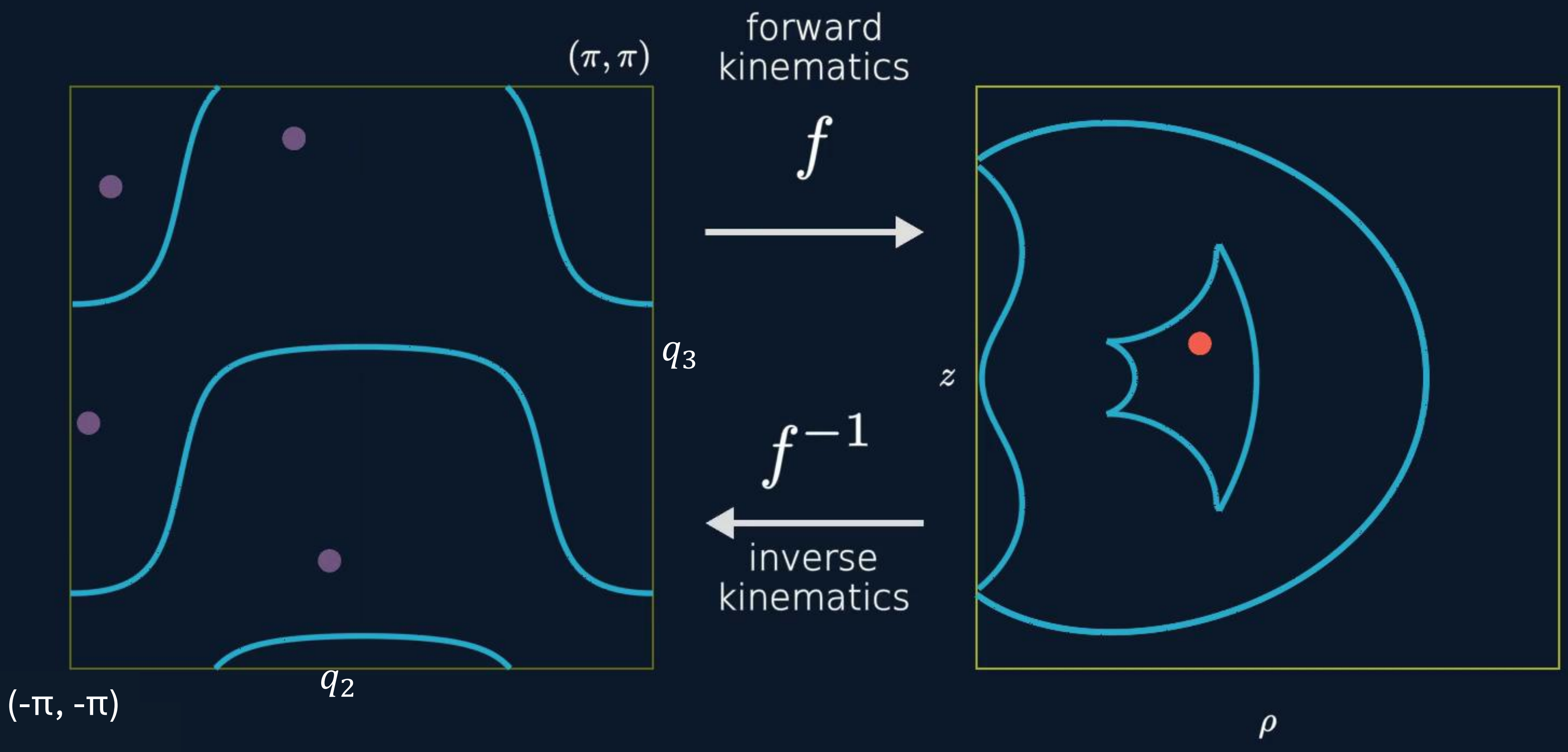


Inverse kinematic solutions are separated by singularities, providing a clear distinction of the paths

Noncuspidal robot

Research activity

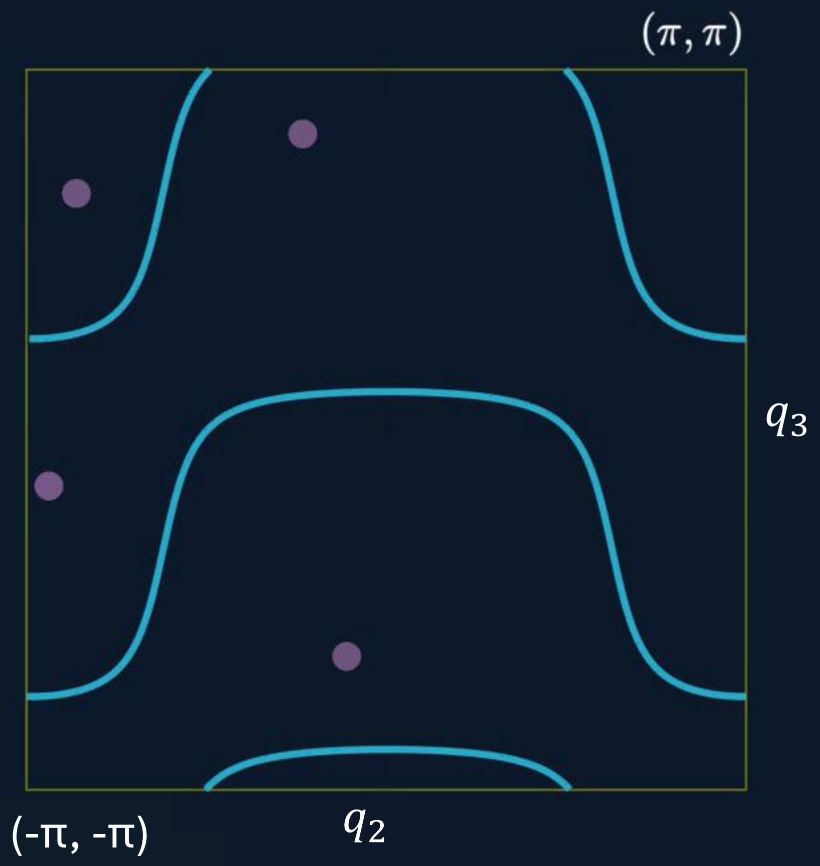
Doctoral work

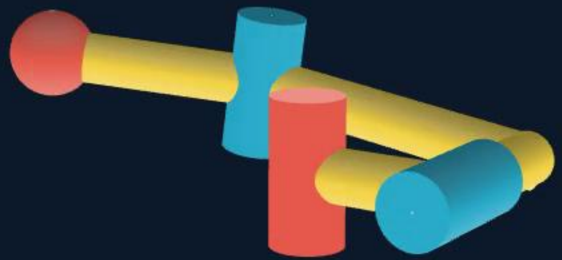


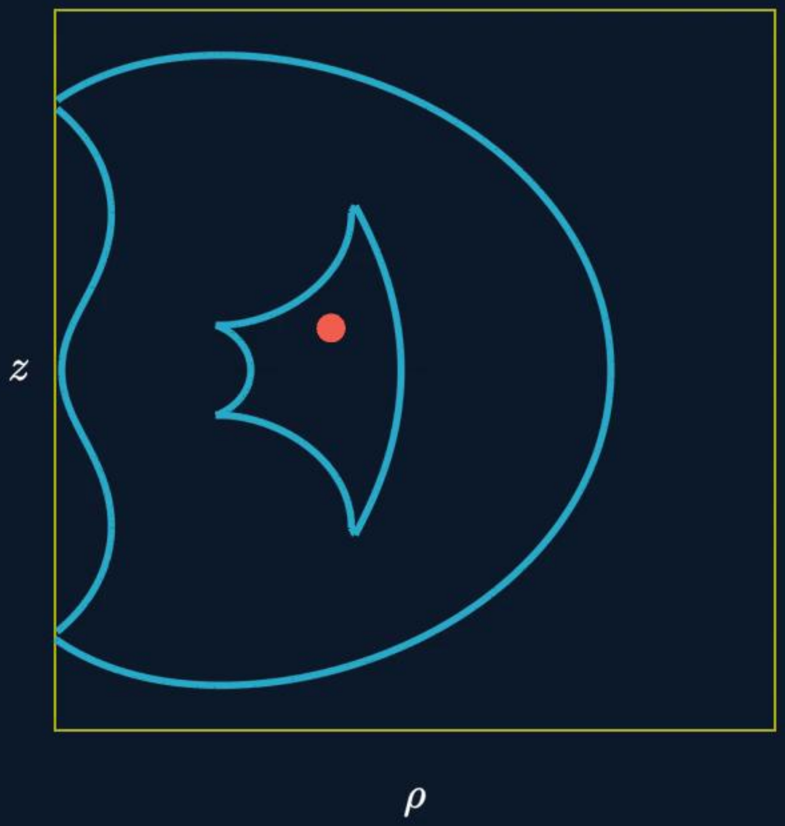
Inverse kinematic solutions are NOT separated by singularities in cuspidal robots

Research activity

Doctoral work





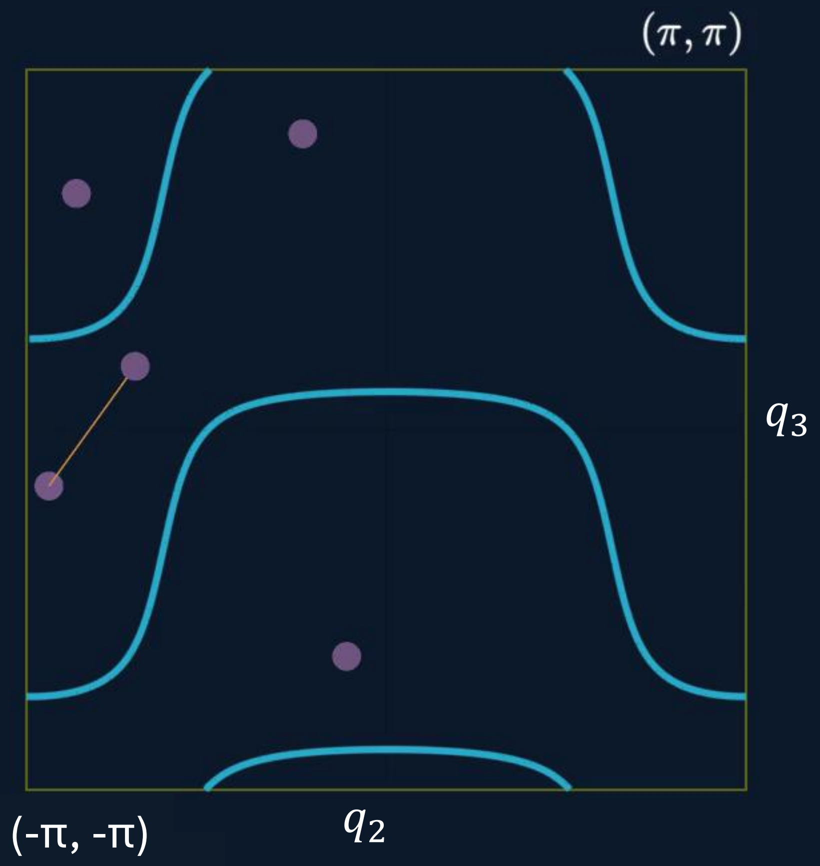


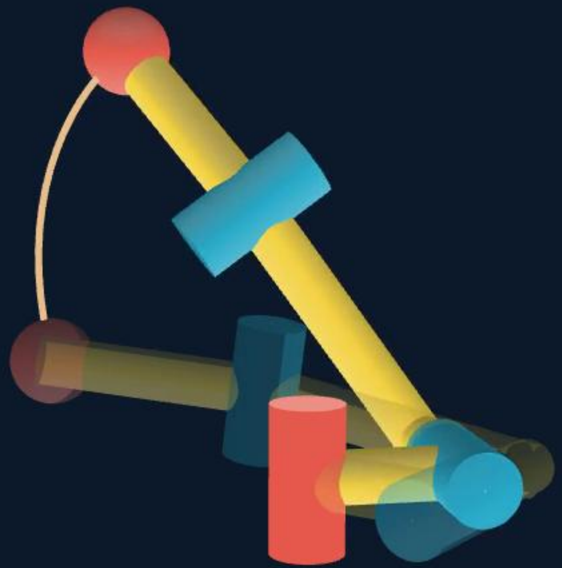
$$\rho = \sqrt{x^2 + y^2}$$

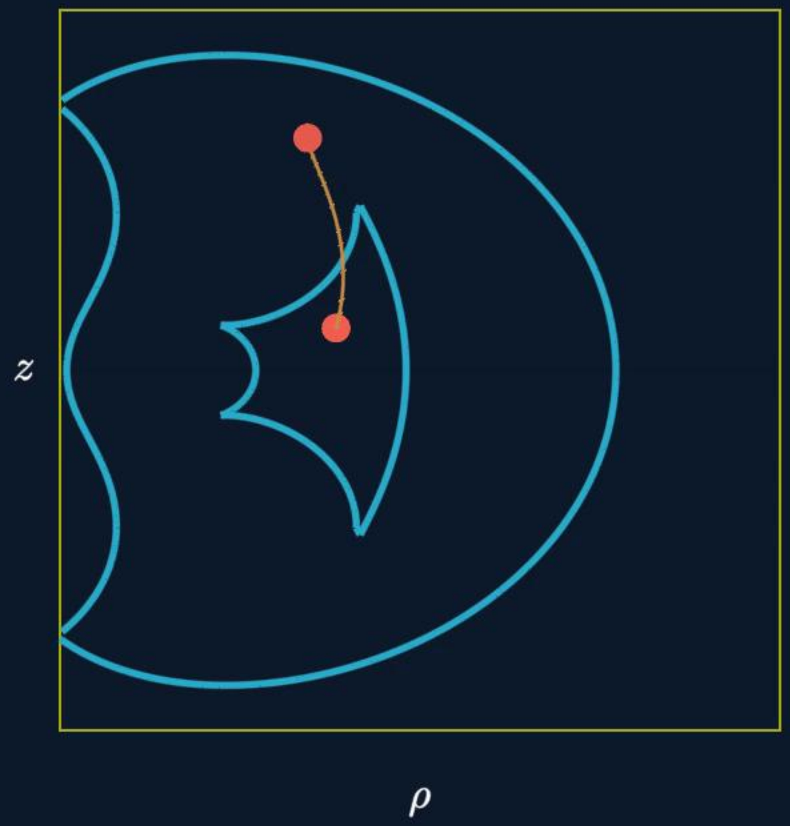
This leads to a change of solution without crossing singularities which causes issues such as infeasible paths and non-repeatable paths

Research activity

Doctoral work







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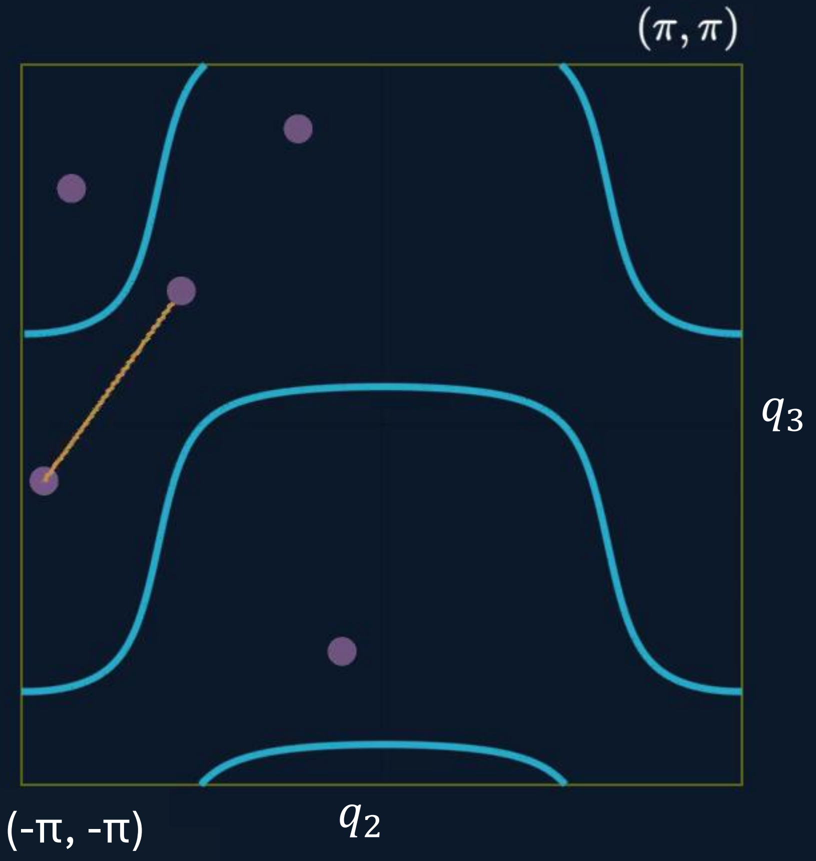

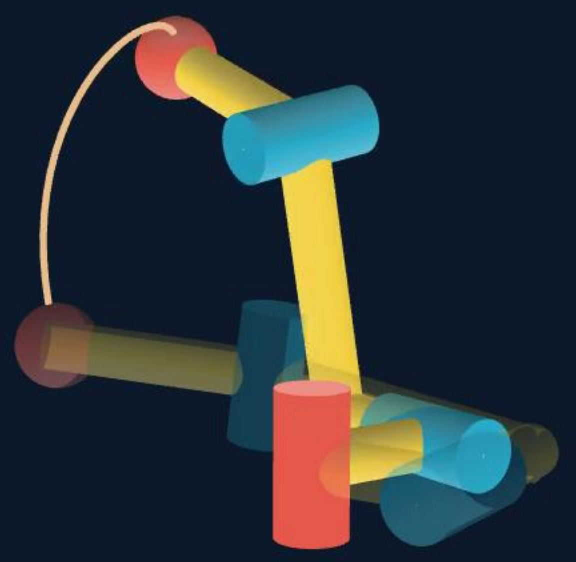

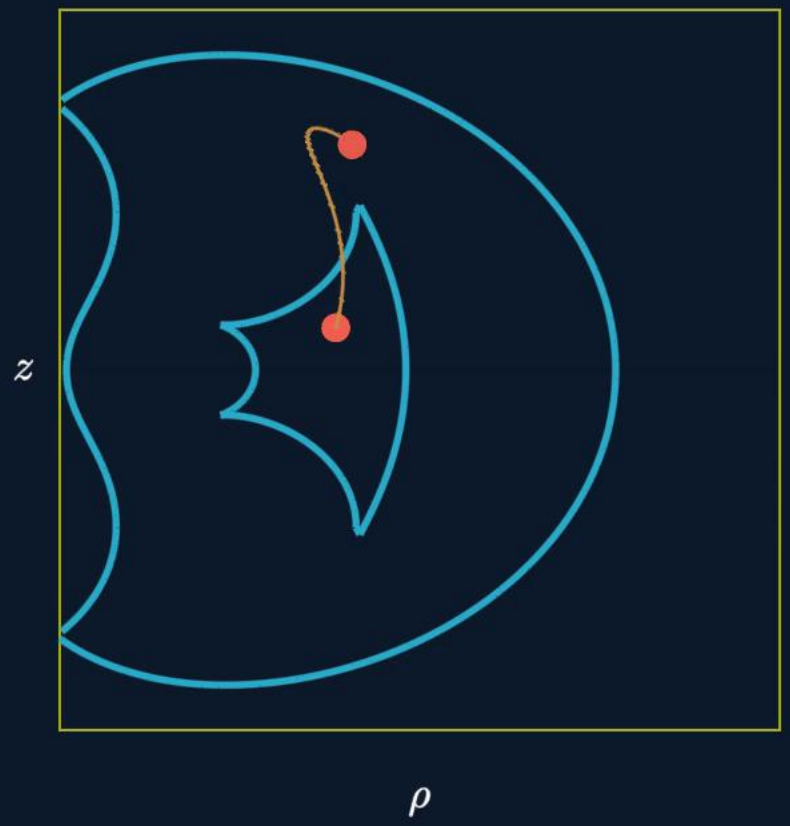

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Research activity

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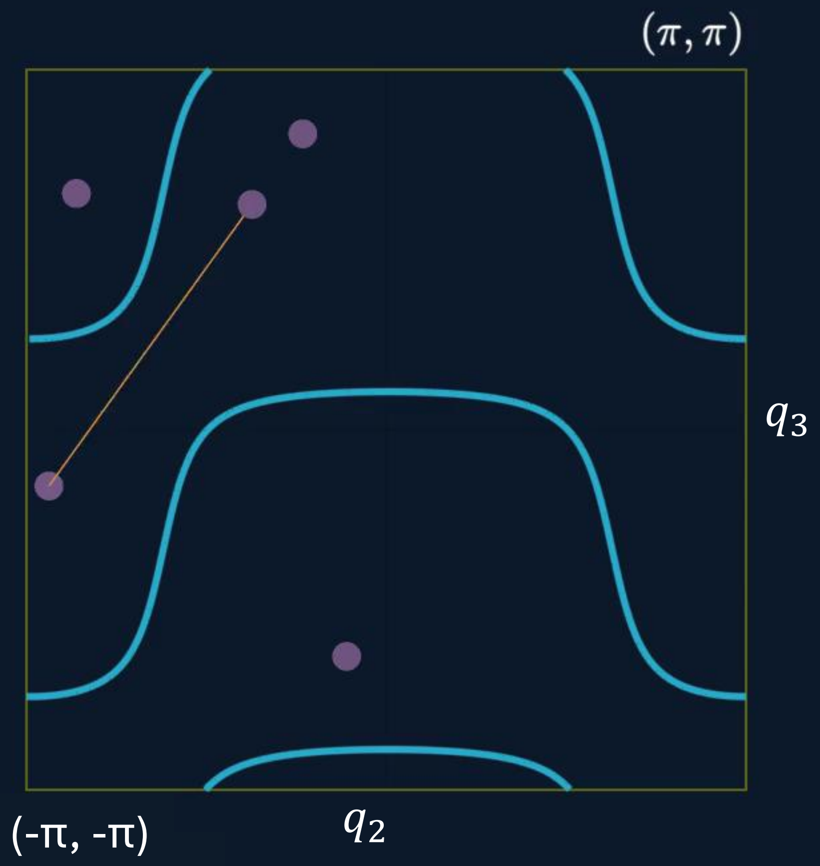

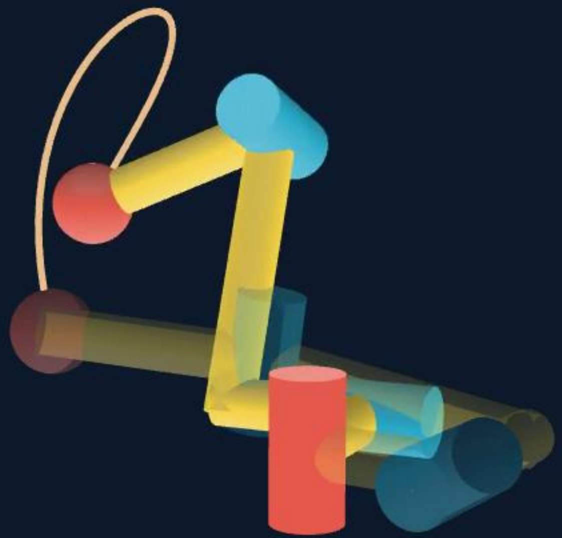

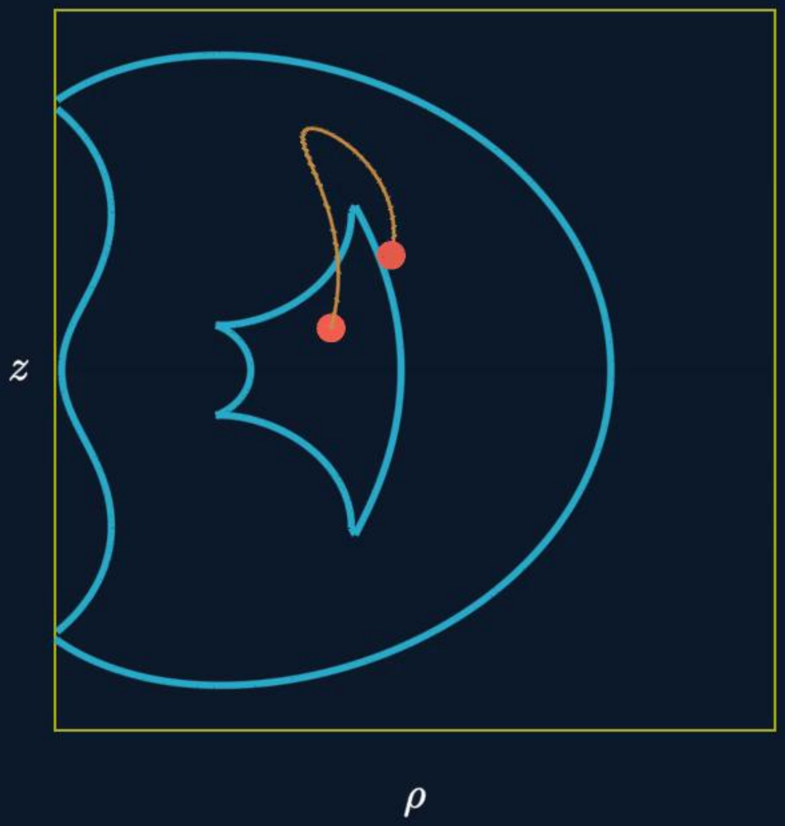

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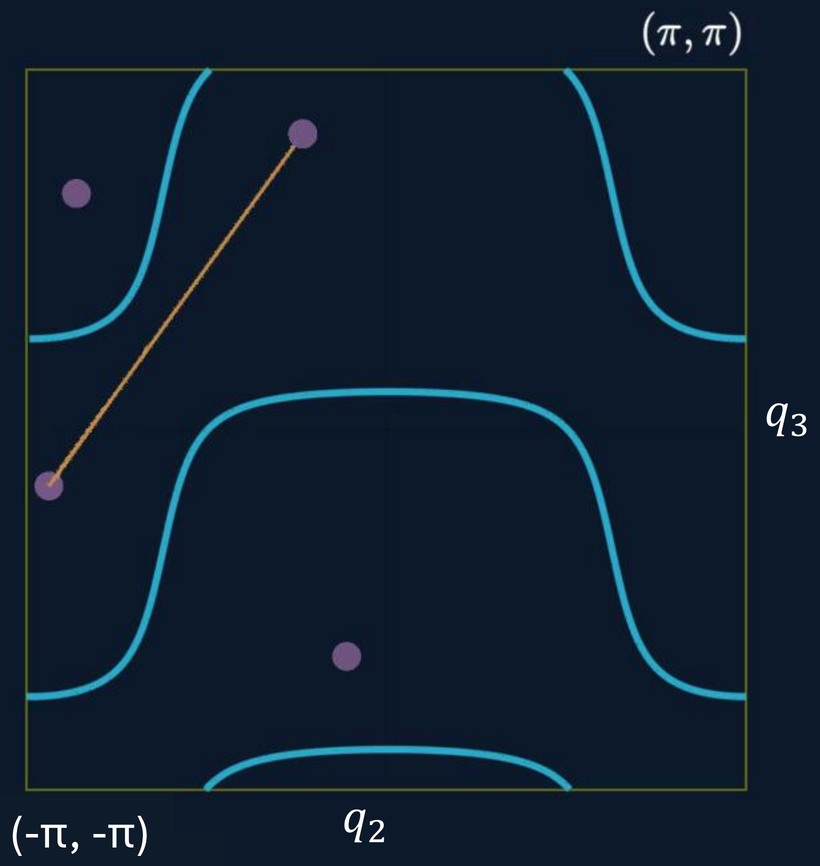
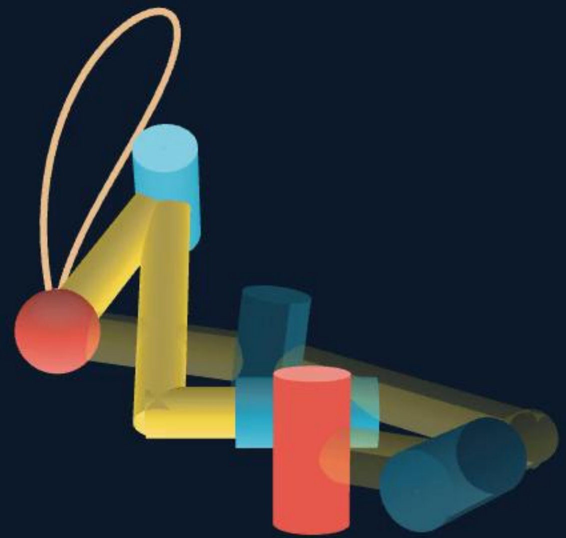
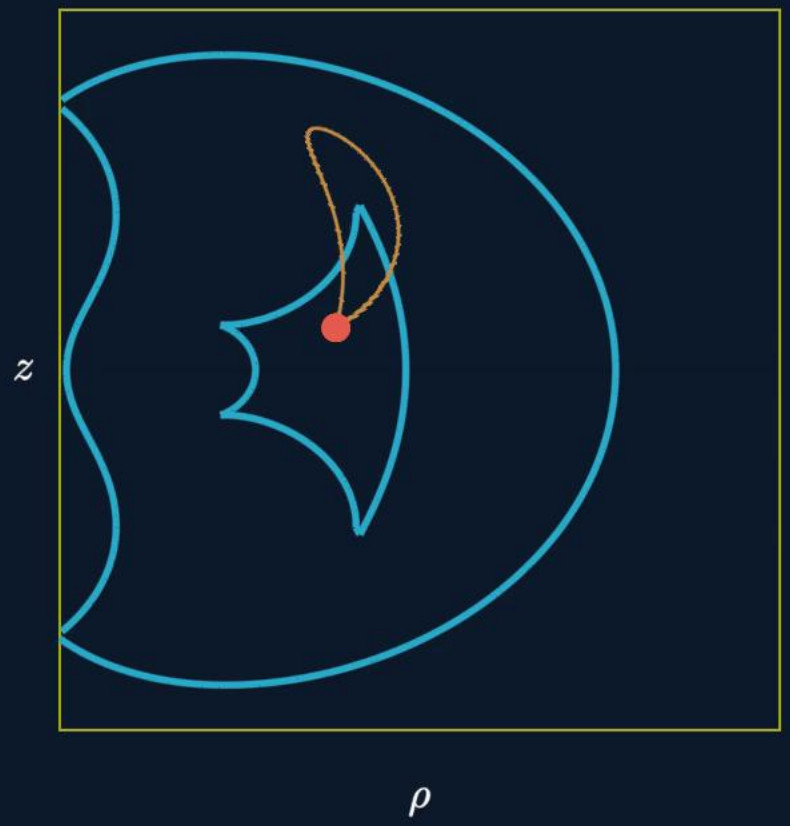
Research activity

Doctoral work





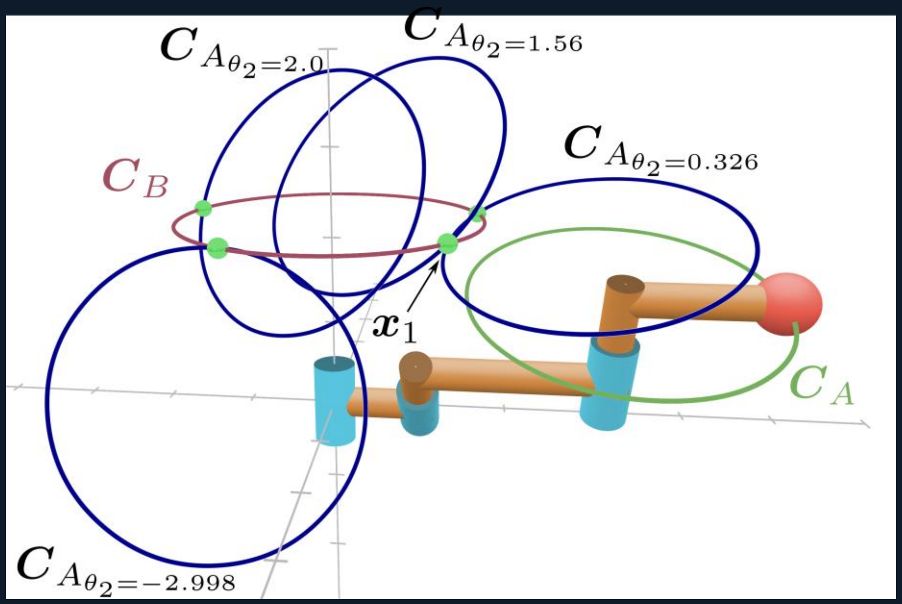


This leads to a change of solution without crossing singularities which causes issues such as infeasible paths and non-repeatable paths

$$\rho = \sqrt{x^2 + y^2}$$

What is Conformal Geometric Algebra?

- CGA is a framework to represent and manipulate:
  - Points, lines, planes, circles, spheres
  - Rigid & conformal transformations (rotate, translate, scale, invert)
- Embeds 3D space into 5D using:
- Two extra basis vectors: e_0 and e_∞
- Combines position + metric info
- Enables distance, angle, and intersection as algebraic operations



Inverse kinematics using CGA

$$X = \mathbf{x} + \frac{1}{2}\|\mathbf{x}\|^2 e_\infty + e_0$$

Conformal embedding of a point

CGA primitives	Direct/primal (OPNS)	Dual (OPNS)
Point $\boldsymbol{p}$	$\boldsymbol{x}$	$\boldsymbol{x}$
Point pair $\boldsymbol{A}$	$\boldsymbol{p}_1 \wedge \boldsymbol{p}_2$	$\boldsymbol{S}_1 \vee \boldsymbol{S}_2 \vee \boldsymbol{S}_3$
Sphere $\boldsymbol{S}$	$\boldsymbol{p}_1 \wedge \boldsymbol{p}_2 \wedge \boldsymbol{p}_3 \wedge \boldsymbol{p}_4$	$\boldsymbol{p}_S - \frac{1}{2}r^2\boldsymbol{e}_\infty$
Plane $\boldsymbol{E}$	$\boldsymbol{p}_1 \wedge \boldsymbol{p}_2 \wedge \boldsymbol{p}_3 \wedge \boldsymbol{e}_\infty$	$\boldsymbol{n} + d\boldsymbol{e}_\infty$
Line $\boldsymbol{L}$	$\boldsymbol{p}_1 \wedge \boldsymbol{p}_2 \wedge \boldsymbol{e}_\infty$	$\boldsymbol{E}_1 \vee \boldsymbol{E}_2$
Circle $\boldsymbol{C}$	$\boldsymbol{p}_1 \wedge \boldsymbol{p}_2 \wedge \boldsymbol{p}_3$	$\boldsymbol{S}_1 \vee \boldsymbol{S}_2$ or $\boldsymbol{S}_1 \vee \boldsymbol{E}_1$

Motivation

Past work

PhD work

Postdoc work

Phase 1
Reliable execution

Phase 2
Skill Scalability

Phase 3
Embodied alignment

Integration

Cuspidal robots

C G A

Summary